Data Warehousing and Data Mining

Part I: Data Mining

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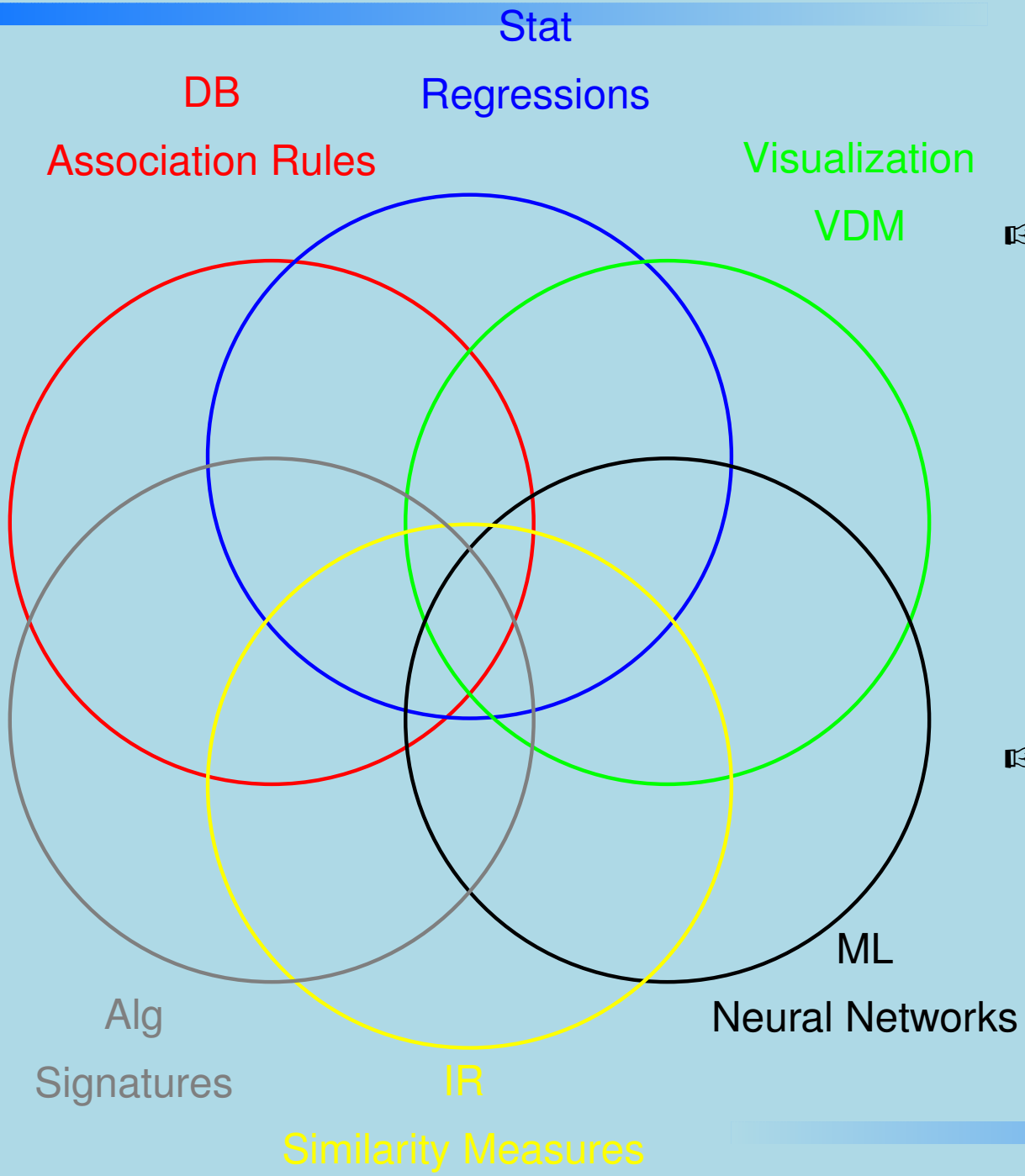
☞ The course material (together with the slides) are available on the Internet:
http://www.inf.unibz.it/dis/teaching/DWDM05/index.html

What is Data Mining?

A few definitions:

- An information extraction activity whose goal is to discover hidden facts contained in databases.
- Nontrivial extraction of implicit, previously unknown and potentially useful information from data, or the search for relationships and global patterns that exist in databases.
- The process of analyzing large amounts of data in order to extract new kinds of useful information (such as implicit relationships between different pieces of information).
- A bunch of algorithms to get potentially interesting information about the dataset

Research Areas Contributed to Data Mining

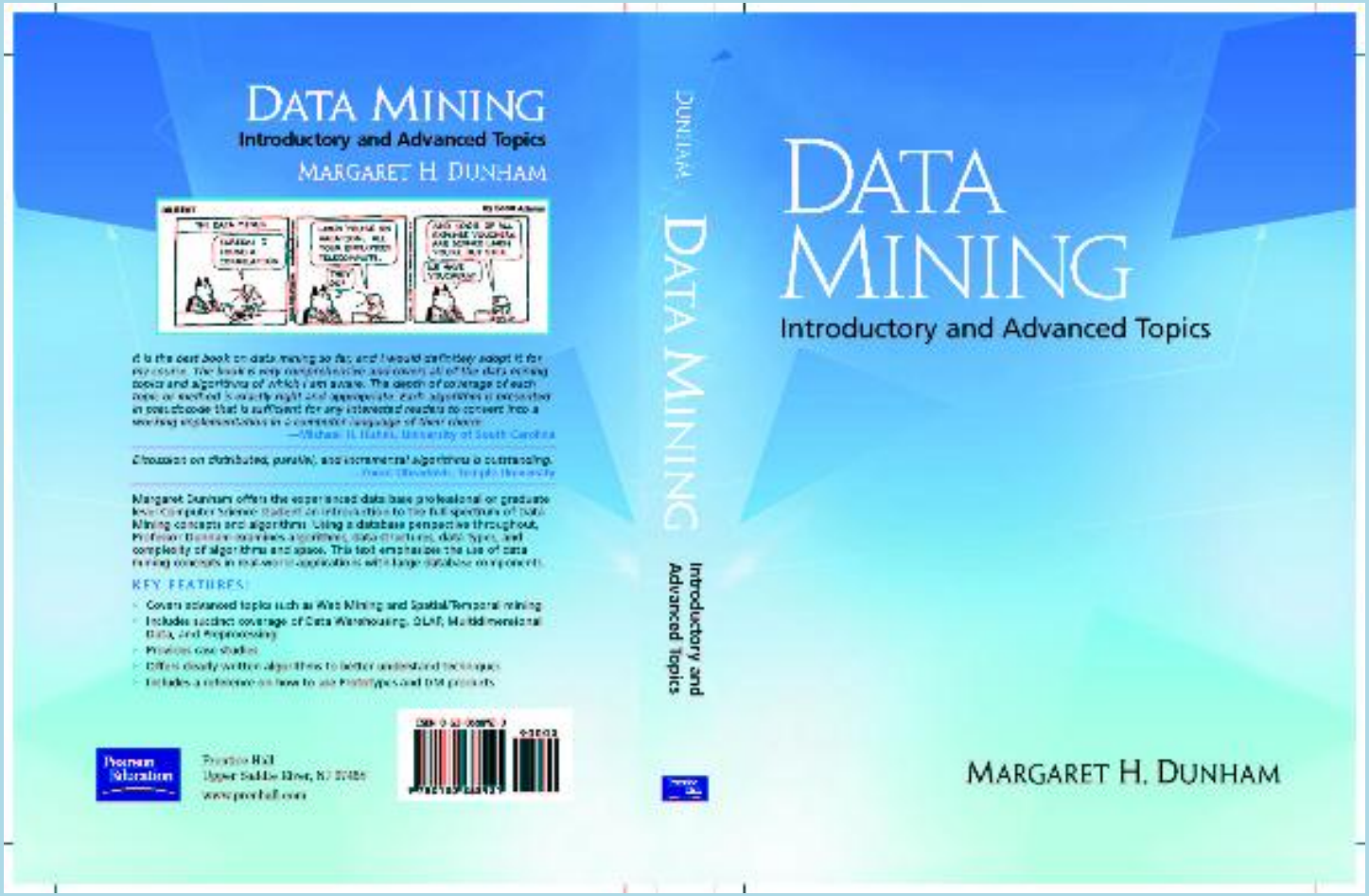


- Lots of research areas contribute to the are of data mining:
  - Databases
  - Statistics
  - Visualization
  - Machine Learning
  - Information Retrieval
  - Algorithms
- I was working in the areas of Databases, Statistics. I've touched a little of IR, and algorithms. ML is my weakest area.

Organization of the Data Mining Part of the Course

- Visual data mining (week 1)
- The core data mining techniques (week 1–3)
  - Classification
  - Clustering
  - Association Rules
- Advanced topics (week 4–6)
  - Web Mining
  - Spatial Mining
  - Temporal Mining

Textbook


DATA MINING
Introductory and Advanced Topics
MARGARET H. DUNHAM
KEY FEATURES:
DUNHAM
DATA MINING
Introductory and Advanced Topics
DATA MINING
Introductory and Advanced Topics
MARGARET H. DUNHAM

Visual Data Mining Techniques

- Scatterplot, scatterplot-matrixes
- Parallel coordinates
- Grand Tour
- Projection pursuit
- Pixel oriented techniques
- Histograms
- APDF, density surface method
- Demo of the 3DVDM system
- Demo of the GGoby system
- Installing the VDM systems
- Demo of the VisDB system

Scatterplots (the Idea)

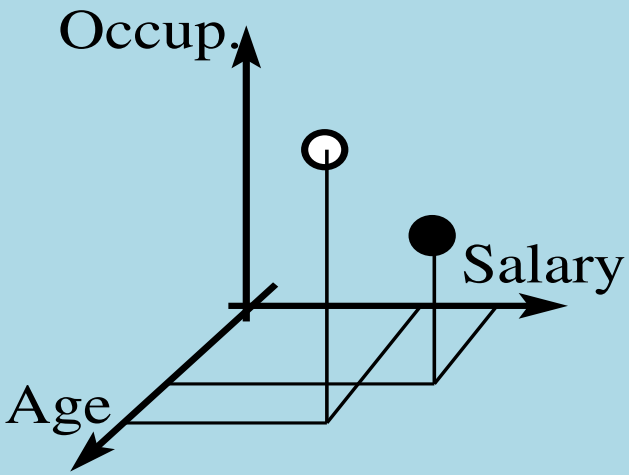
- Scatterplots visualize database points as points in the Euclidean vector space (with perpendicular axes).
- Each database tuple is treated as a point in the high dimensional vector space.
- In essence, the attributes of the tables to the visual properties of the visual world (X,Y,Z,shape,color, ...) (Such the mapping is also called OPM (Object Property Model).

Age	Salary	Occupation	Sex
35	1'500	Driver	F
55	2'500	Clerk	M
...	...	...	...

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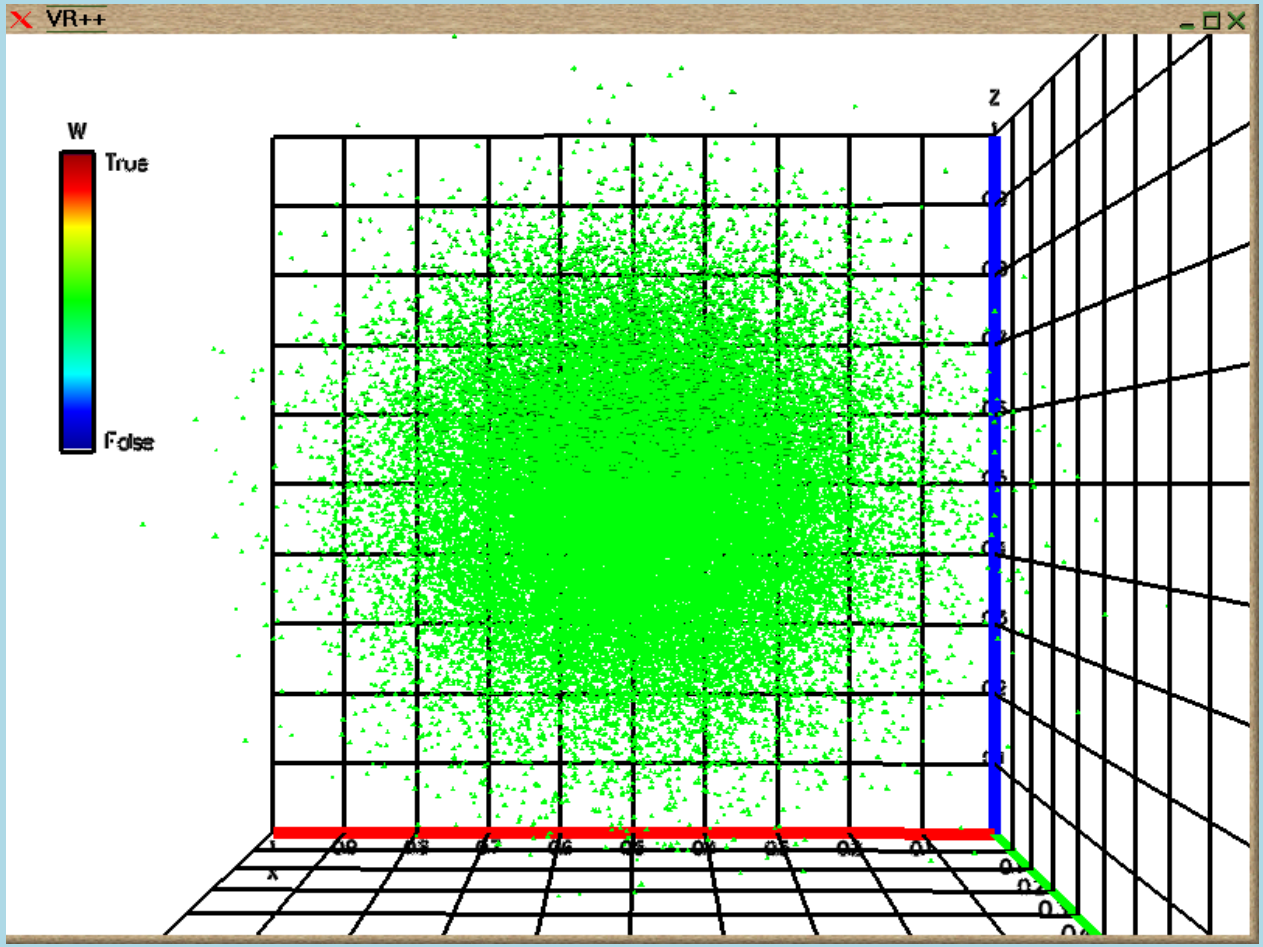
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Scatterplots (A Two-Dimensional Example)

Age	Salary	Height
35	1’500	1.58
65	2’300	1.88
25	1’000	1.70
60	2’000	1.80
30	1’200	1.50
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Scatterplots (Demo of the 3DVDM System)



- 3DVDM/scatterplot
- `art-ball`
- `art-2sphere`
- `art-cone`
- `art-cube-several-spl`
- `art-cube-sphere`
- `art-line4d`
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- `art-spiral`

Scatterplot-matrices (the Idea)

- Scatterplot-matrices is used to visualize high dimensional data with the number of dimensions more than 2 or 3.
- Scatterplot-matrices are visualized in three steps:

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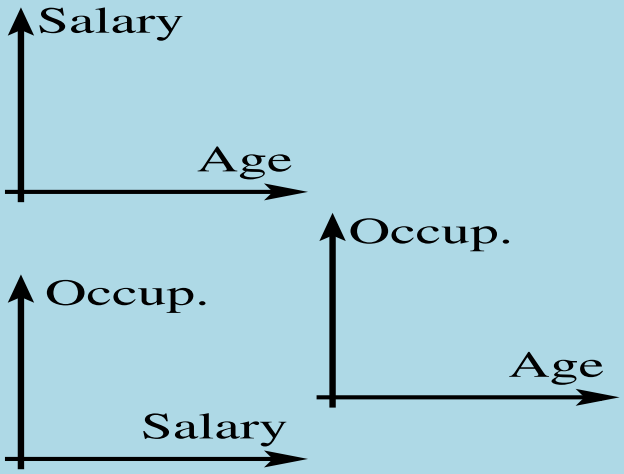
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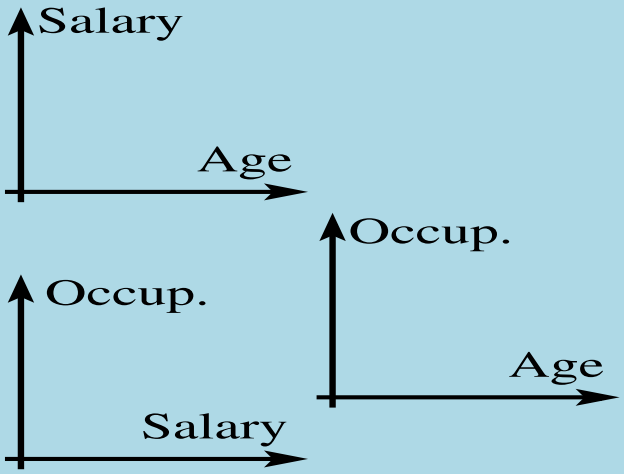
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  - Generate the input database into all possible projections of two attributes
  - Visualize all two-attribute tables as scatterplot
  - Arrange the visualized scatterplots in a matrix

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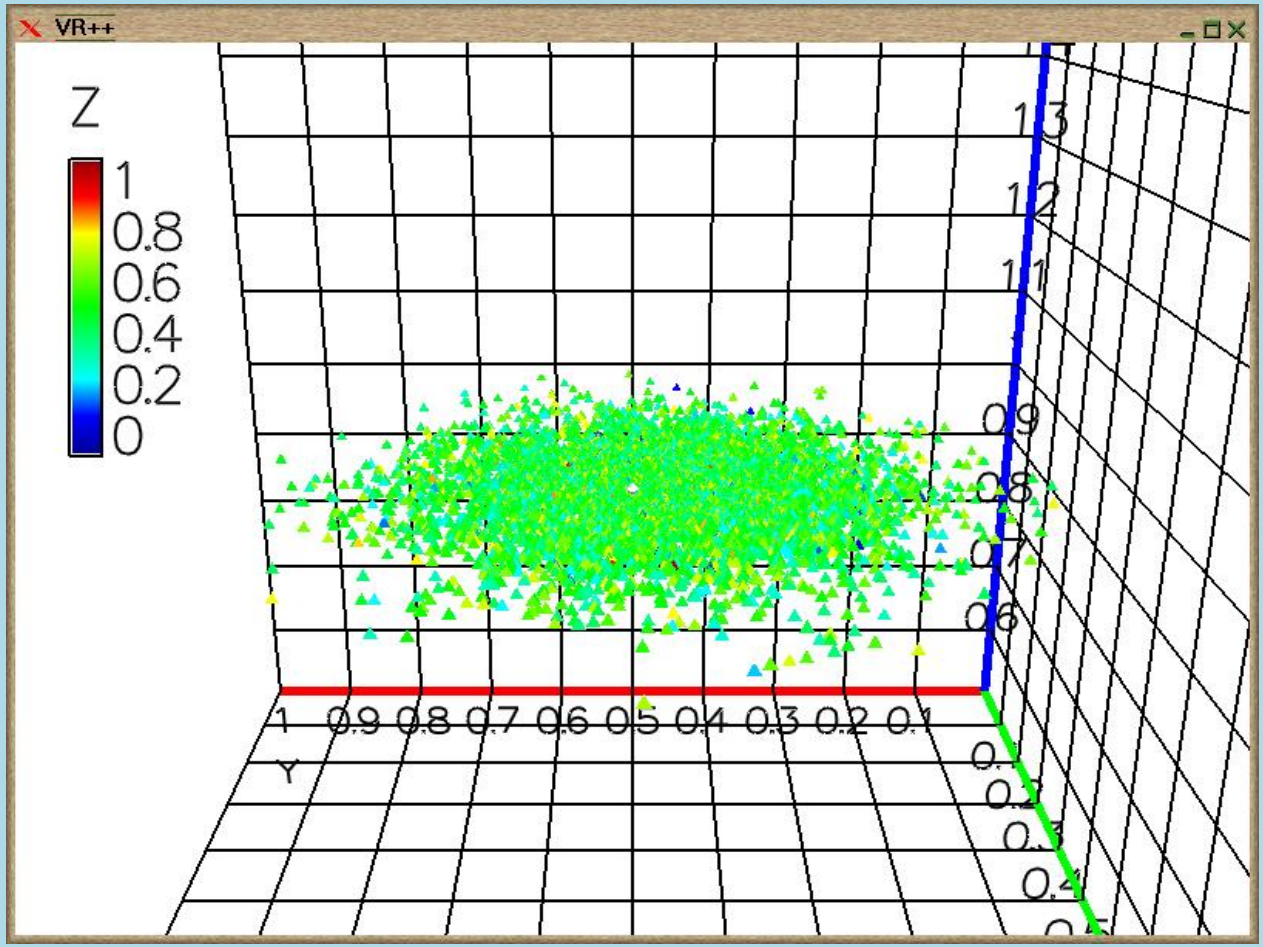
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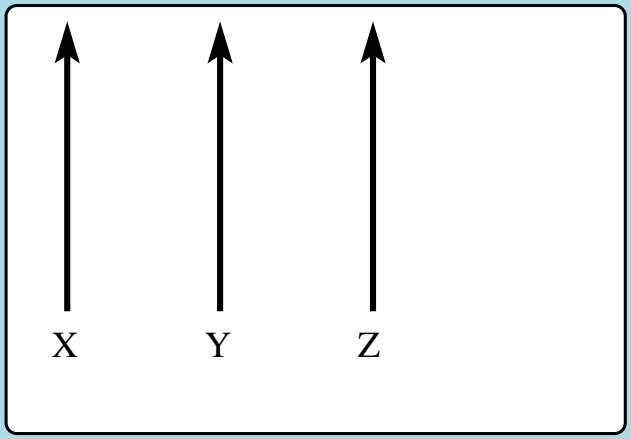
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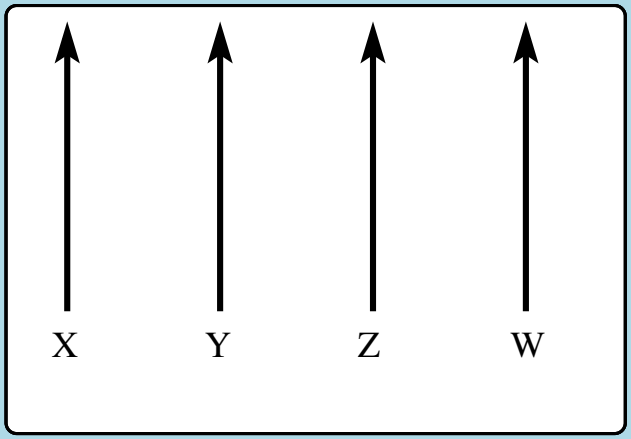
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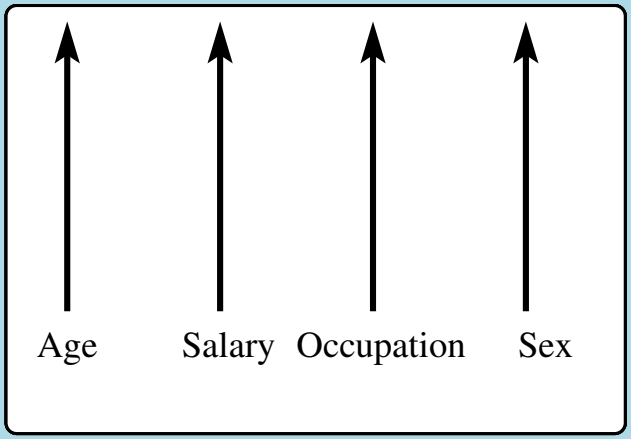
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- ☞ A point in these coordinates is visualized in per axe bases.

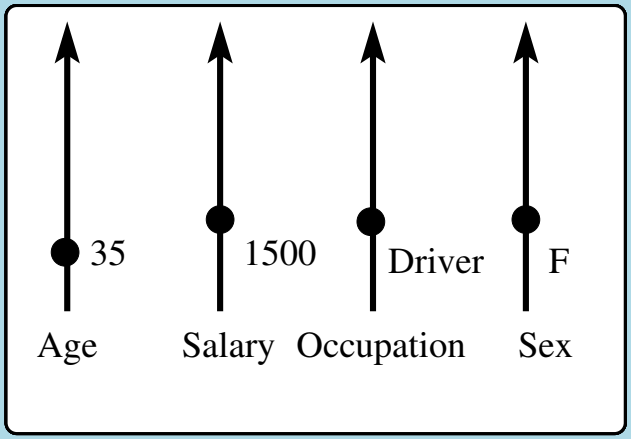


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- It is very easy to increase the dimensionality of the visualization as needed
- A point in these coordinates is visualized in per axe bases.
- The appropriate points are connected with lines

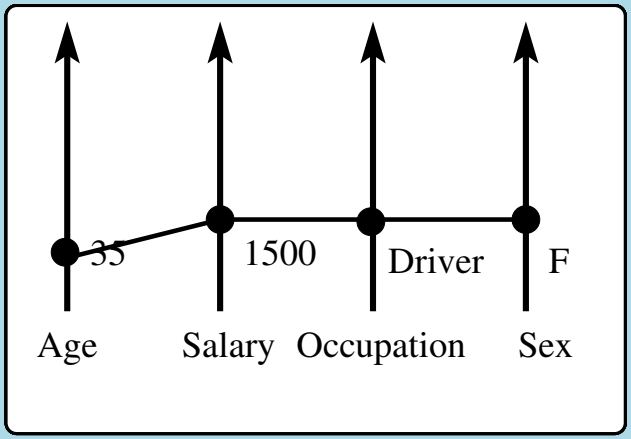


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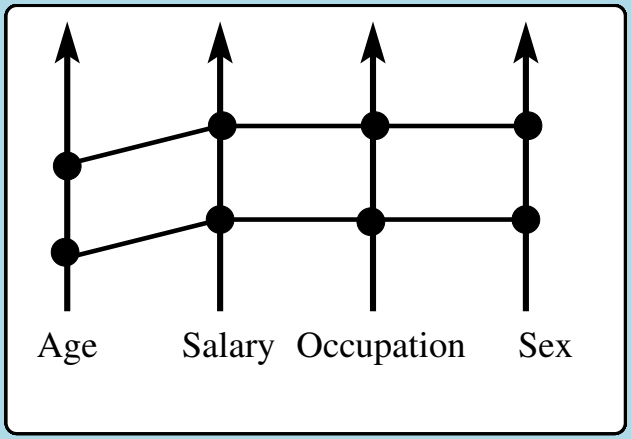


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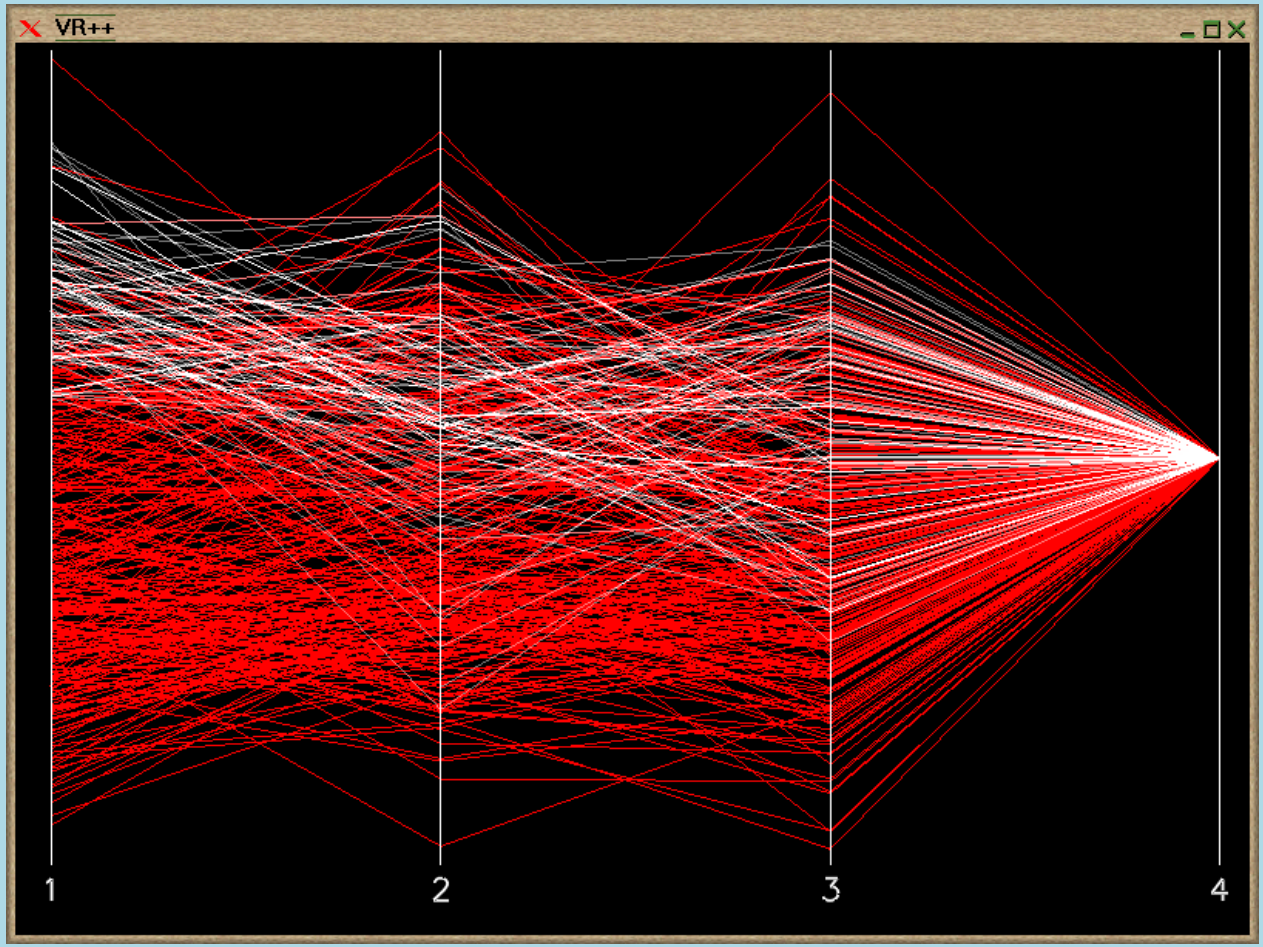


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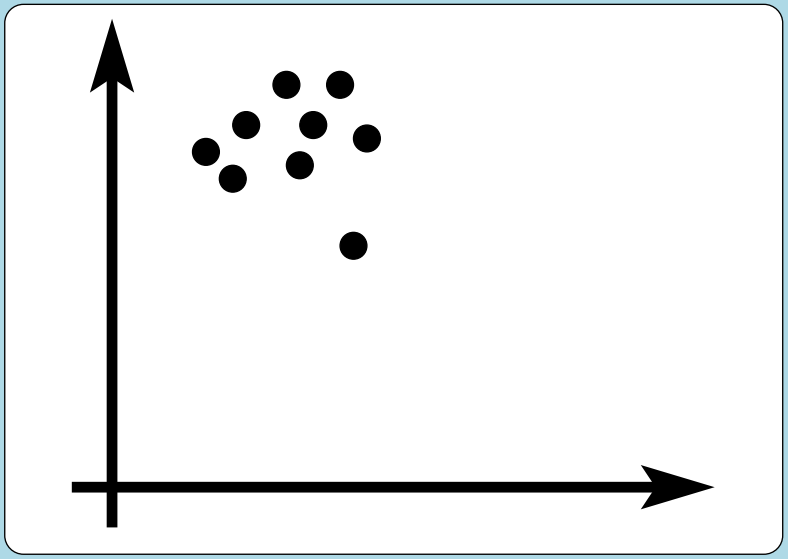
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Grand Tour (the Idea, $d = 1$)

- ☞ Grand tour: is a random sequence of projections from n dimensional data into low dimensional space d ($d = 1, 2, 3$)

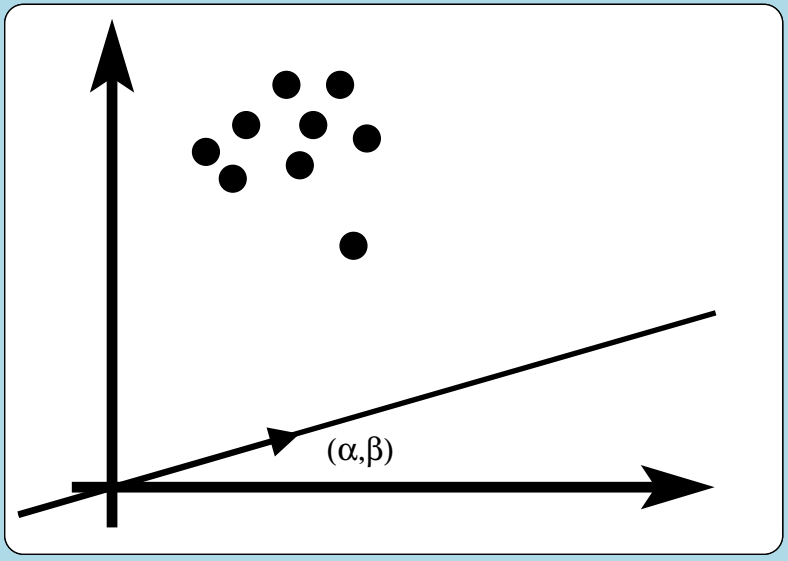
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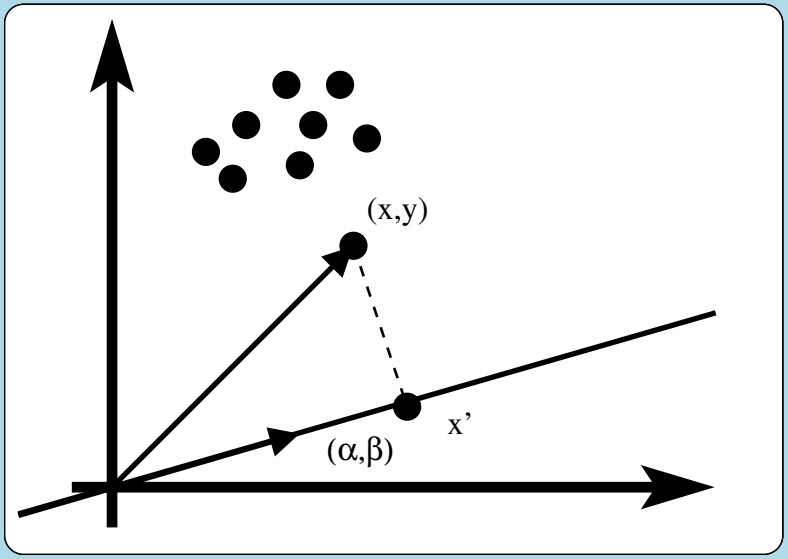
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  2. Project the database points (x, y) onto the line:

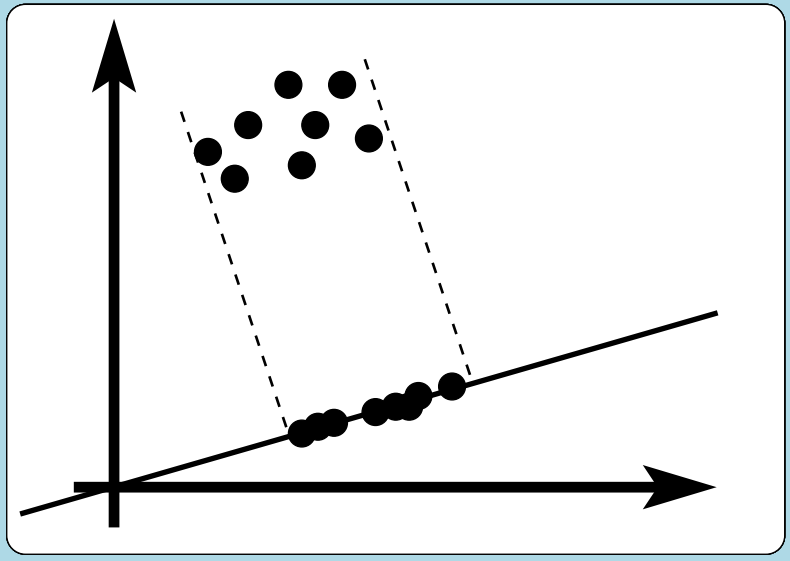
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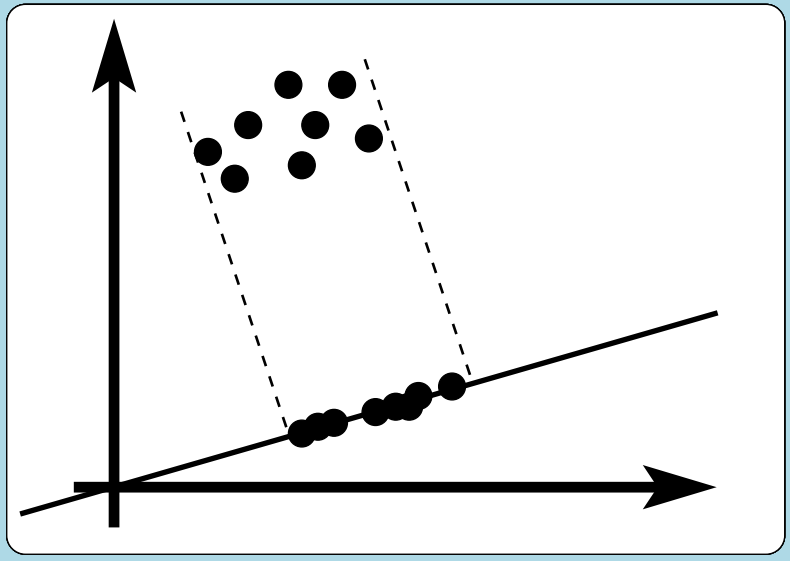


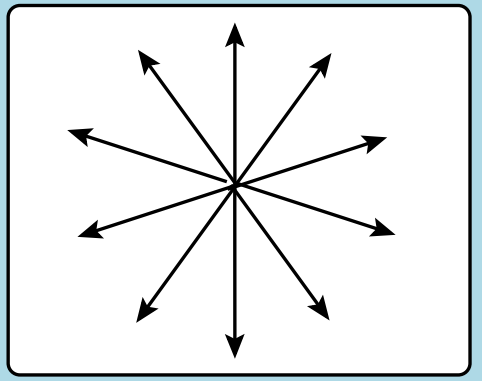
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  3. Iterate 1–2 for all other directions in a continuous dynamic sequence



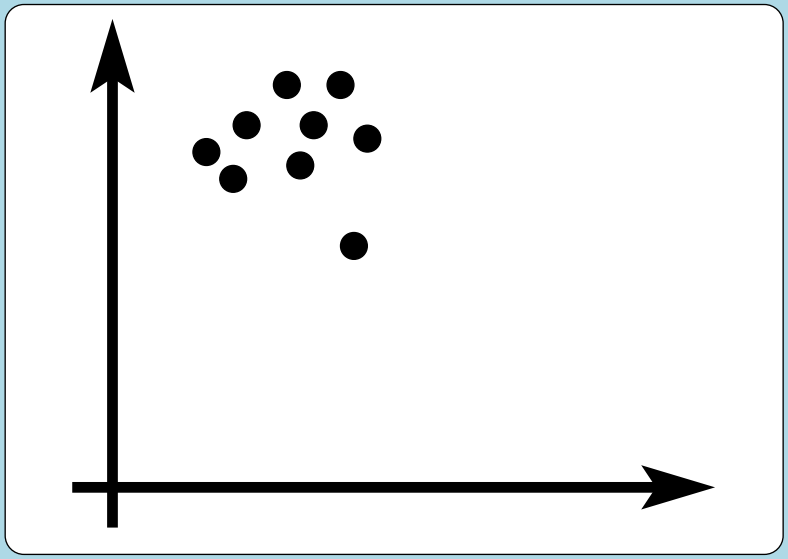


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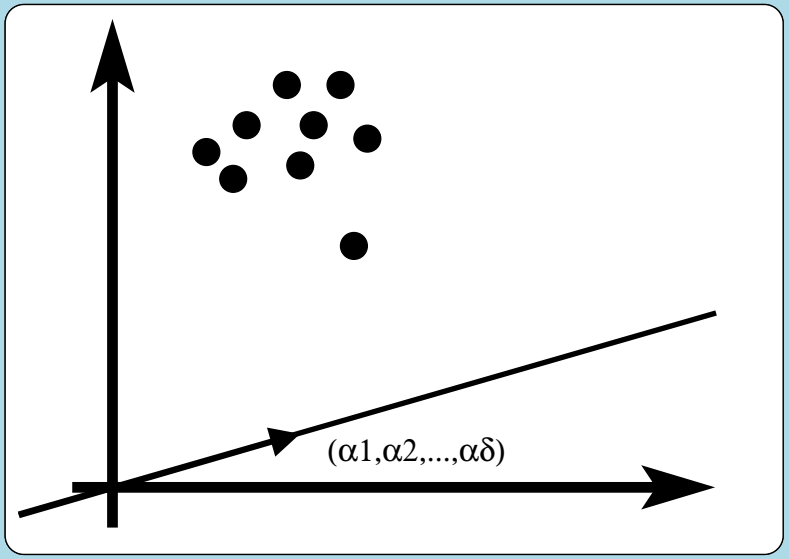


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1. Chose the projection direction

$$(\alpha_1, \ldots, \alpha_d) = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \ldots & \alpha_{1d} \\ \alpha_{21} & \alpha_{22} & \ldots & \alpha_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \ldots & \alpha_{nd} \end{pmatrix}$$

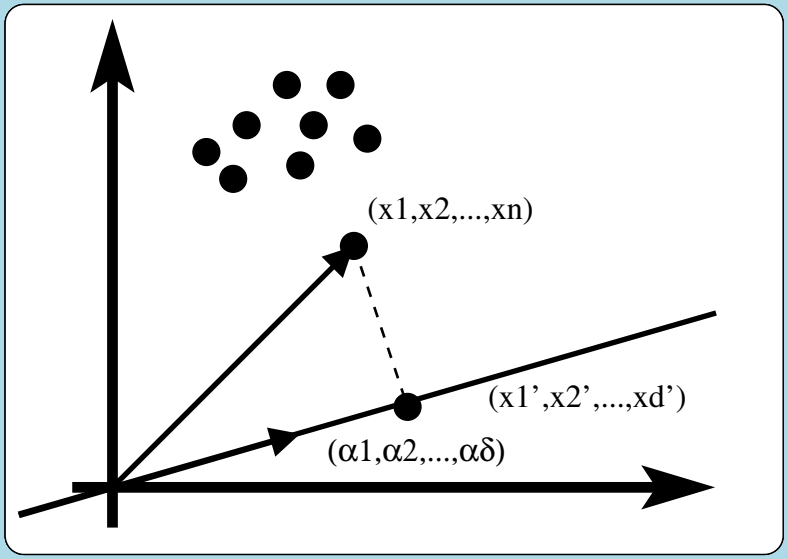


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2. Project database point $(x_1, x_2, \ldots, x_n)$ on to the chosen projection:

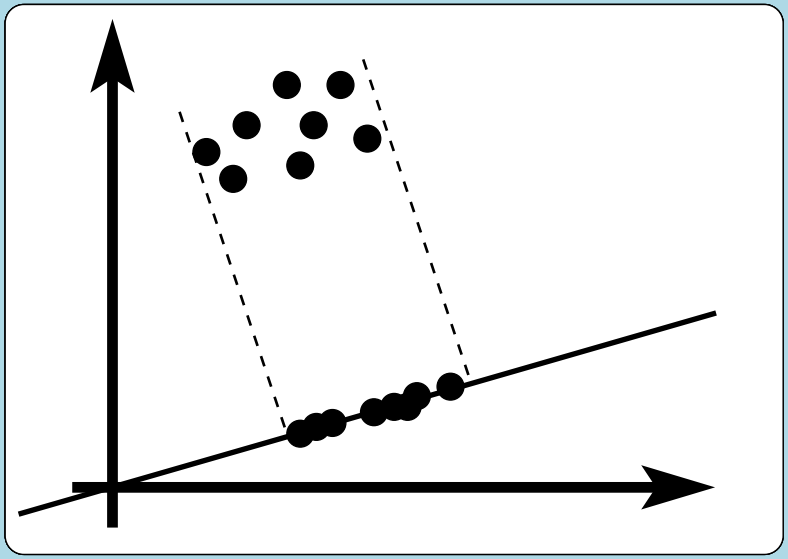
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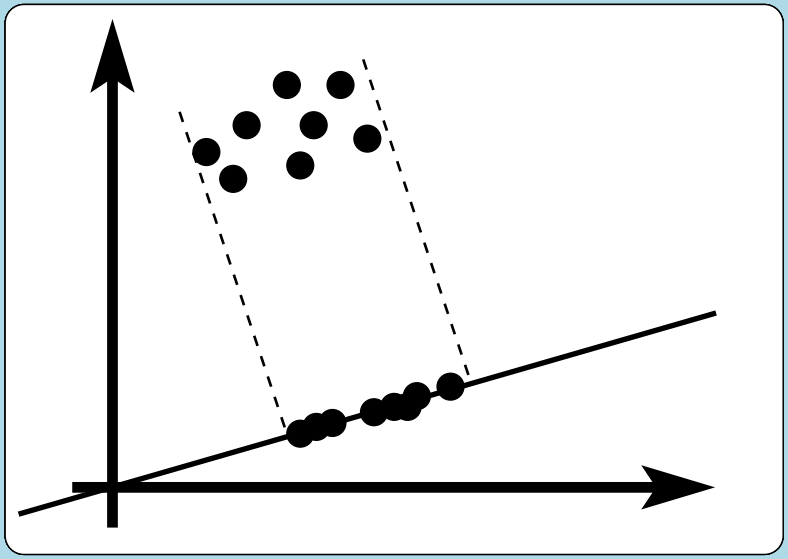
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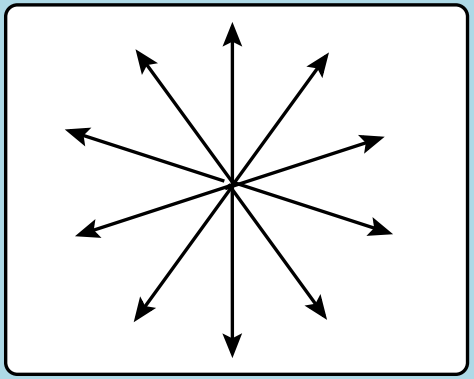
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Grand Tour (Example)

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60	2'000	1.80
30	1'200	1.50
55	1'800	1.80
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- $n = 3,$
- $d = 2,$
-

$$(\alpha_1, \alpha_2) = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

Grand Tour (Demo)





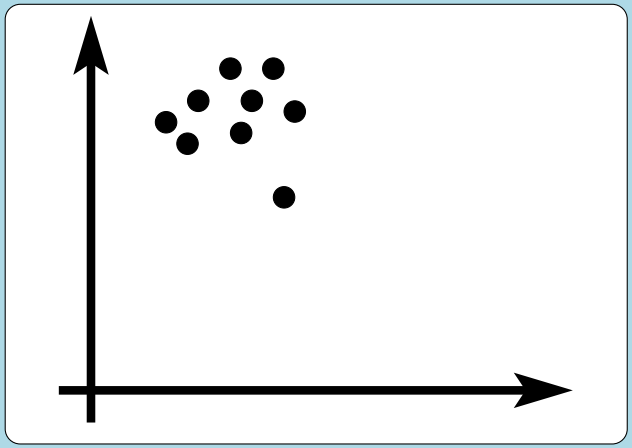
ggoby, laser dataset

Projection Pursuit (the Idea Only)

- ☞ Projection pursuit “improves” grand tour by:
  - Shows only those low dimensional that expose interesting patterns

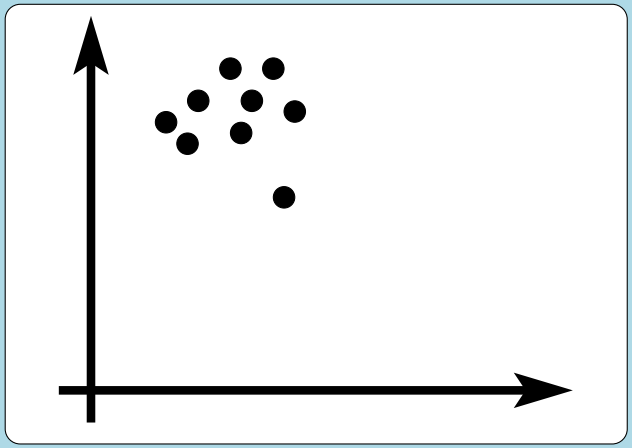
Projection Pursuit (the Idea Only)

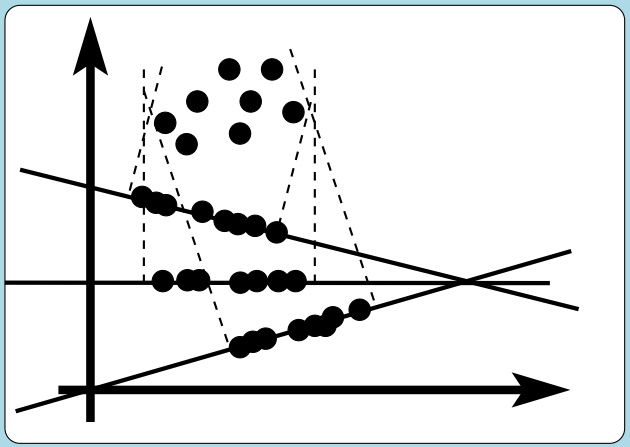
- Projection pursuit “improves” grand tour by:
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- Projection pursuit is computed and visualized in three steps:



Projection Pursuit (the Idea Only)

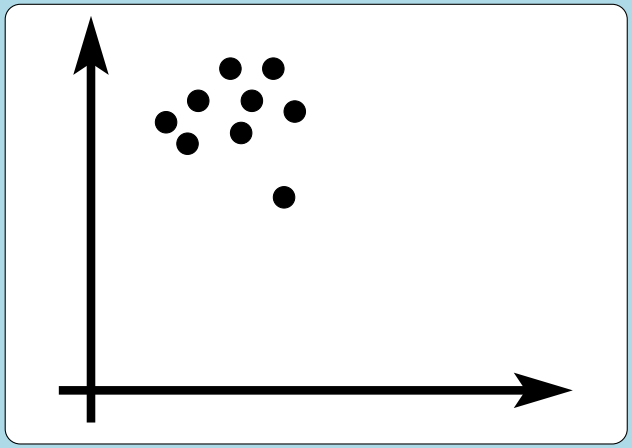
- Projection pursuit “improves” grand tour by:
  - Shows only those low dimensional that expose interesting patterns
- Projection pursuit is computed and visualized in three steps:
  - Select all possible low projections of high dimensional space
  - Compute the criteria function function (projection pursuit index) for each projection

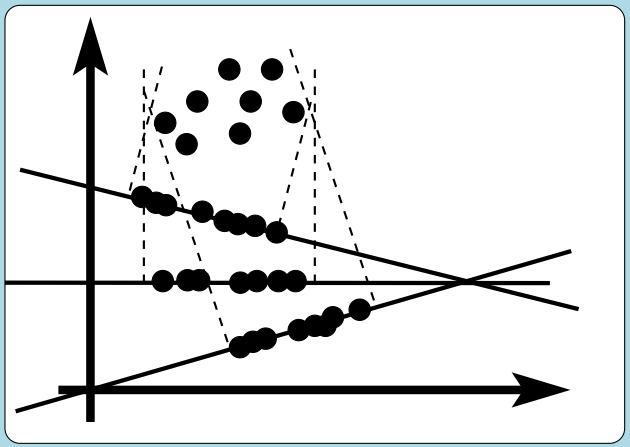


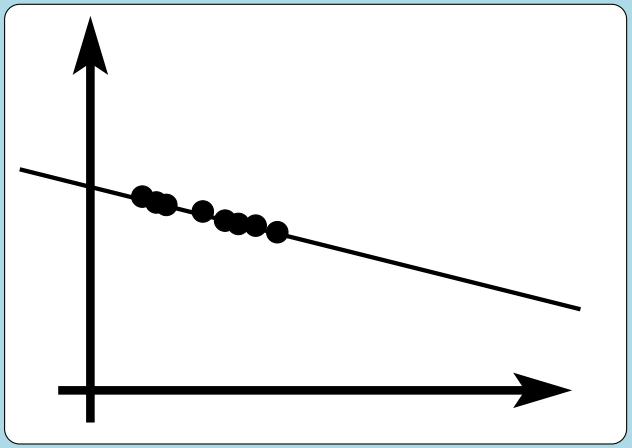


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  - Visualize the “best” projections only

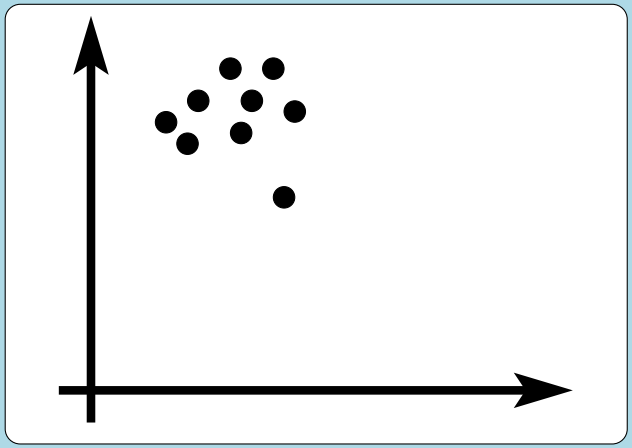


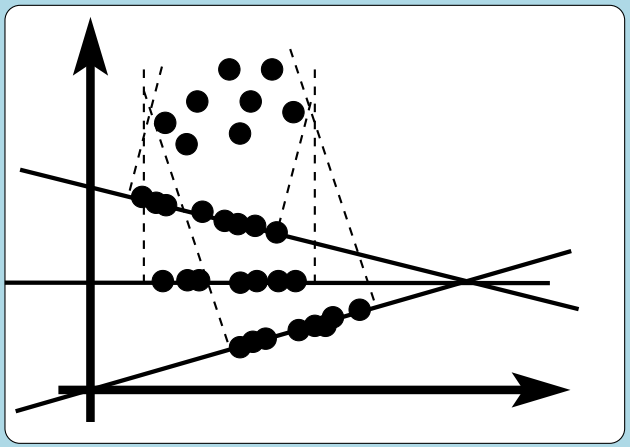


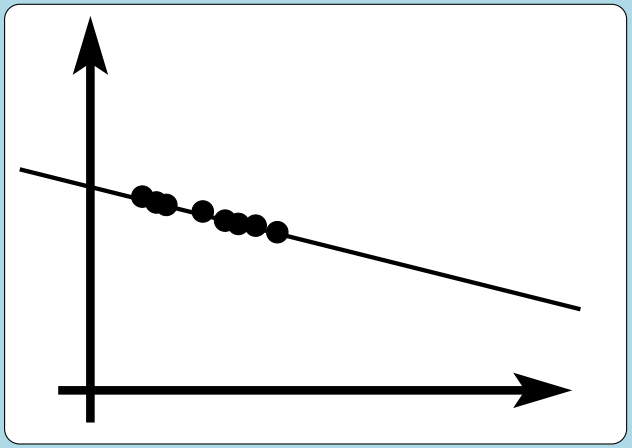


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  - Select all possible low projections of high dimensional space
  - Compute the criteria function function (projection pursuit index) for each projection
  - Visualize the “best” projections only
- A good criterion function, which compares the distribution of the points in the projection with the normal distribution. The further the distributions are away from each other the better it is.







Projection Pursuit + Grand Tour (the Idea Only)

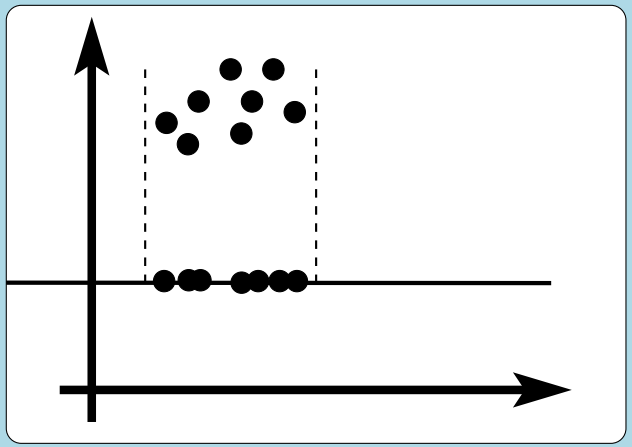
- Projection pursuit combines the grand tour and projection pursuit by addressing shortcoming of the methods:

Projection Pursuit + Grand Tour (the Idea Only)

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  1. Start with a random projection

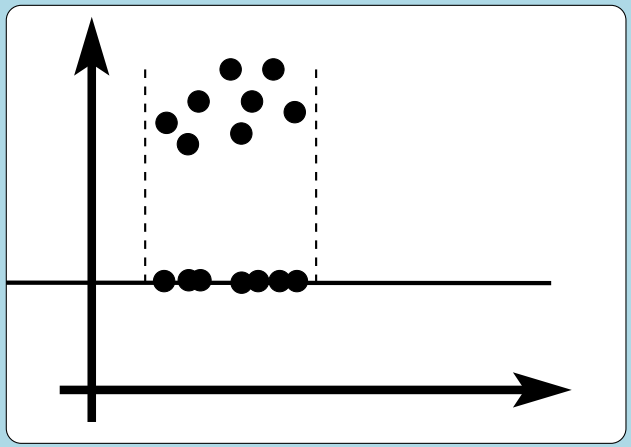
Projection Pursuit + Grand Tour (the Idea Only)

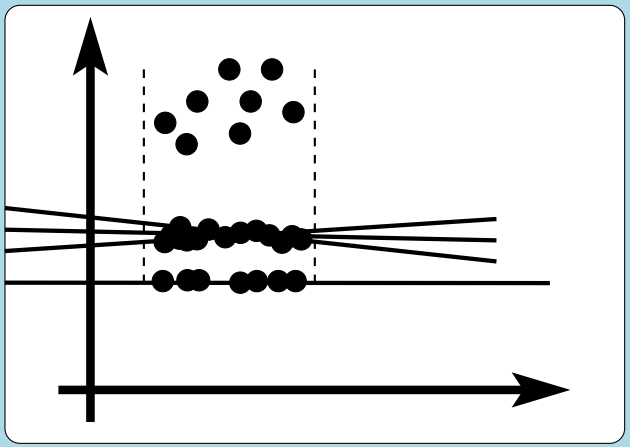
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Projection Pursuit + Grand Tour (the Idea Only)

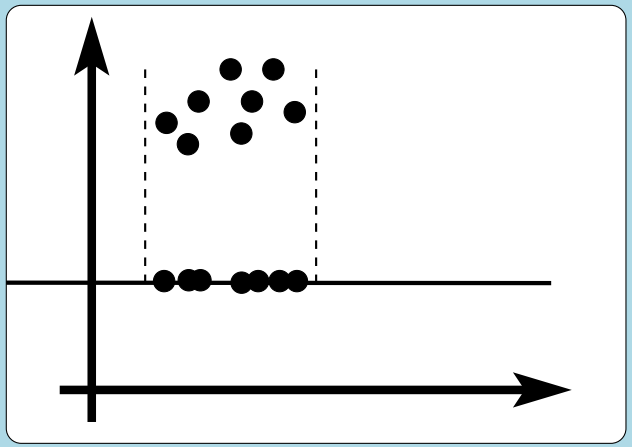
- Projection pursuit combines the grand tour and projection pursuit by addressing shortcoming of the methods:
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  3. Compute the best projection (or the direction of the next best projection) B over the set of projections P

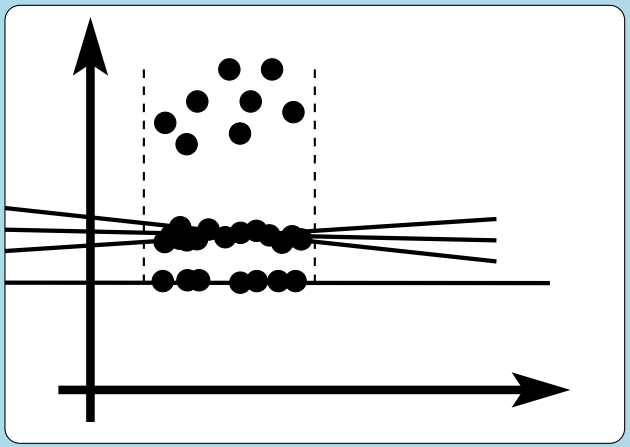


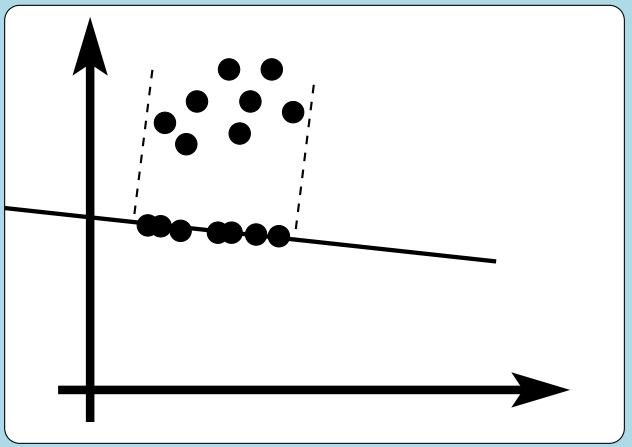


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  1. Start with a random projection
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  4. Animate from the initial projection to the local best projection B

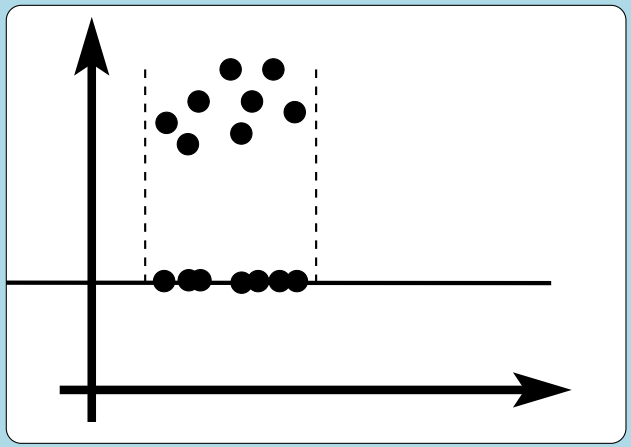


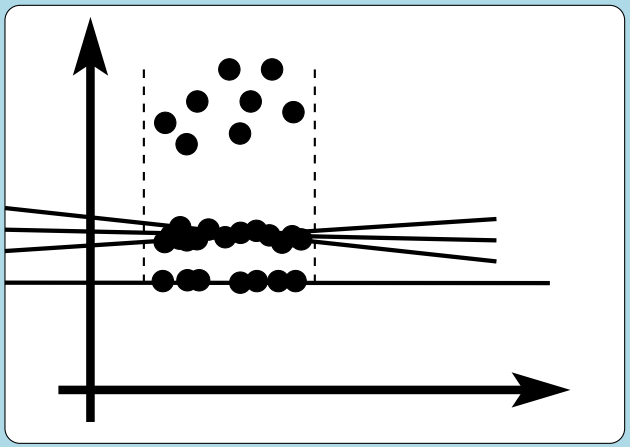


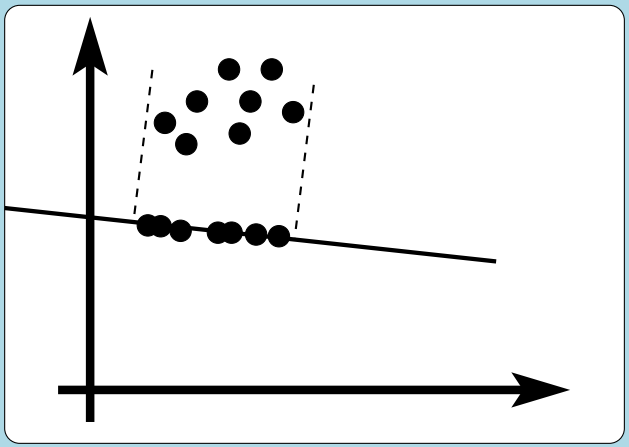


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  4. Animate from the initial projection to the local best projection B
  5. Iterate steps 1–3 until no better projections are found

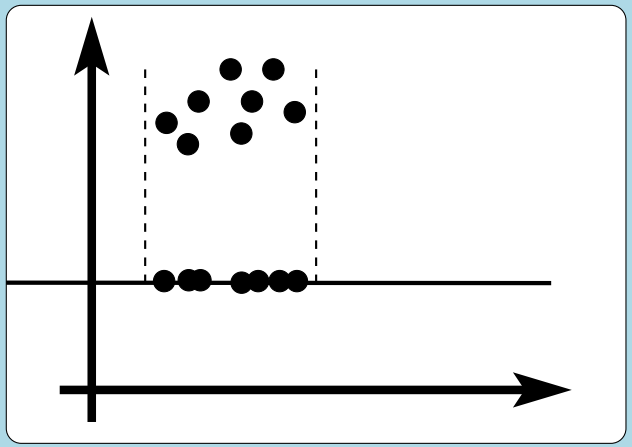


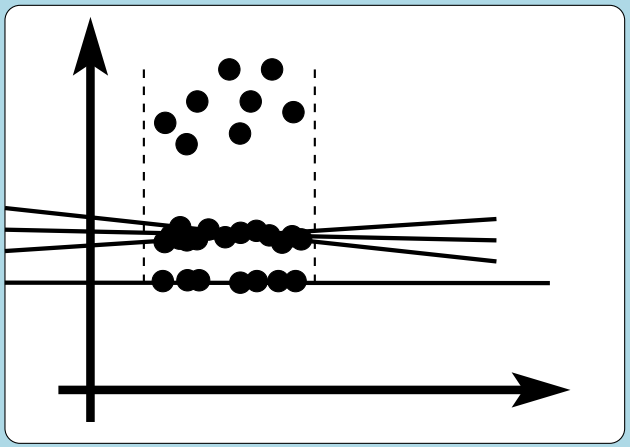


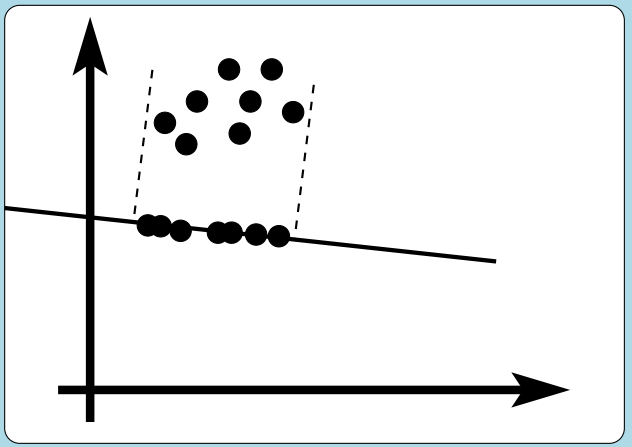


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  4. Animate from the initial projection to the local best projection B
  5. Iterate steps 1–3 until no better projections are found
  6. Reset the initial projection to a random one; iterate 1–5.







Dense Pixel Displays (the Idea)

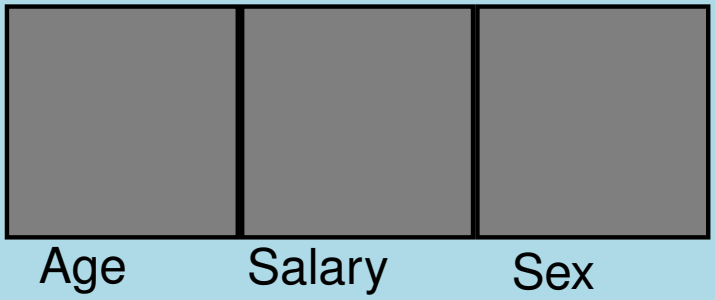
☞ The idea is to use each pixel of the screen to visualize the data items.

Age	Salary	Sex
35	1’500	F
55	2’500	M
...	...	...

Dense Pixel Displays (the Idea)

- The idea is to use each pixel of the screen to visualize the data items.
- Each attribute is visualized in a separate window (One tuple is visualized in several windows).

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35	1’500	F
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...	...	...



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- The idea is to use each pixel of the screen to visualize the data items.
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35	1’500	F
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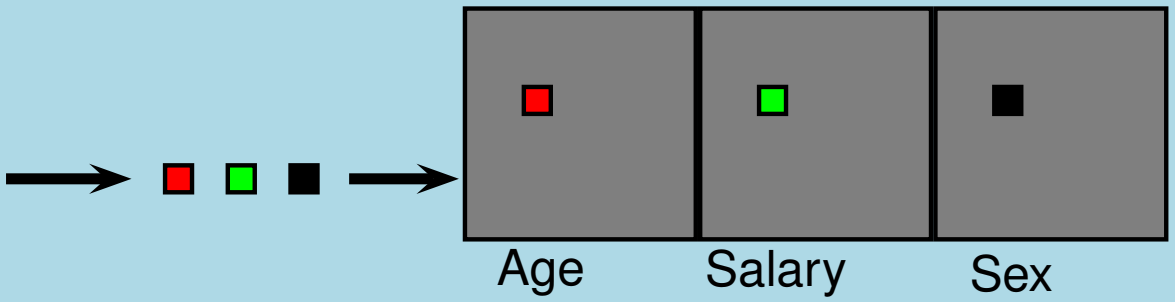




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- The idea is to use each pixel of the screen to visualize the data items.
- Each attribute is visualized in a separate window (One tuple is visualized in several windows).
- Each value for each attribute is assigned with a color
- The color pixels are visualized at the same coordinates of coordinate windows

Age	Salary	Sex
35	1’500	F
55	2’500	M
...	...	...



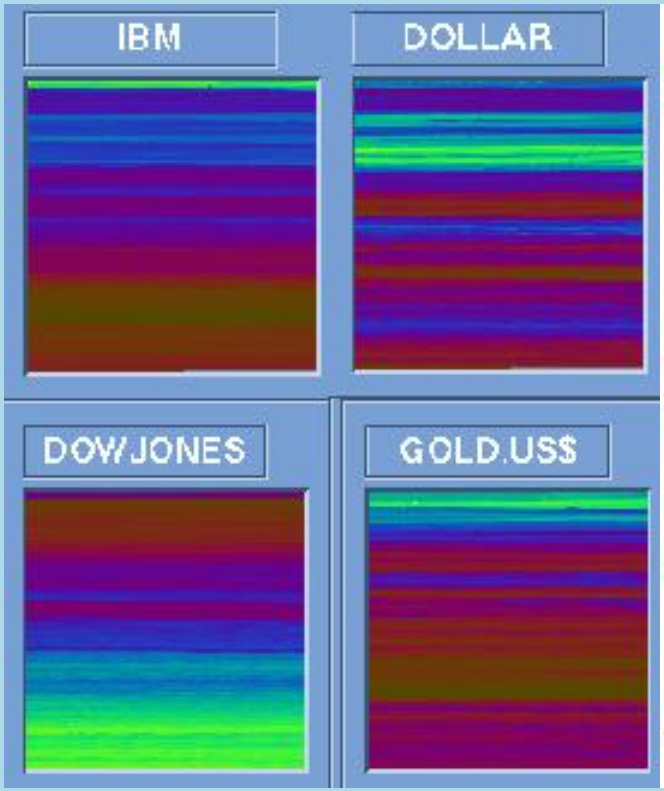
Dense Pixel Displays (Example I)

☞ Time series of stock data:

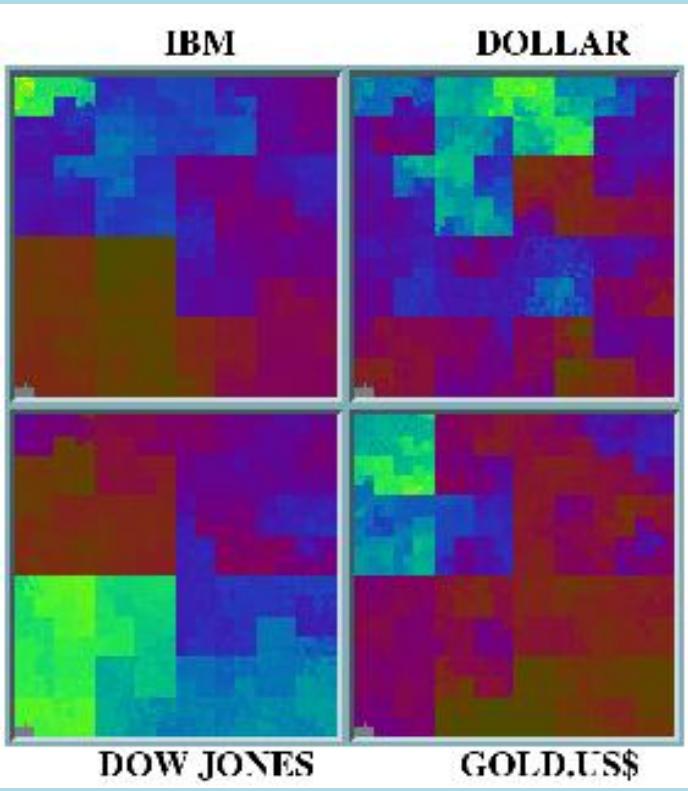
Time	IBM	DOLLAR	DOW JONES	GOLD US$
12.12.98 12:01	7.8	3.8	11567	56.80
...	...	...	...	...

Dense Pixel Displays (Example I)

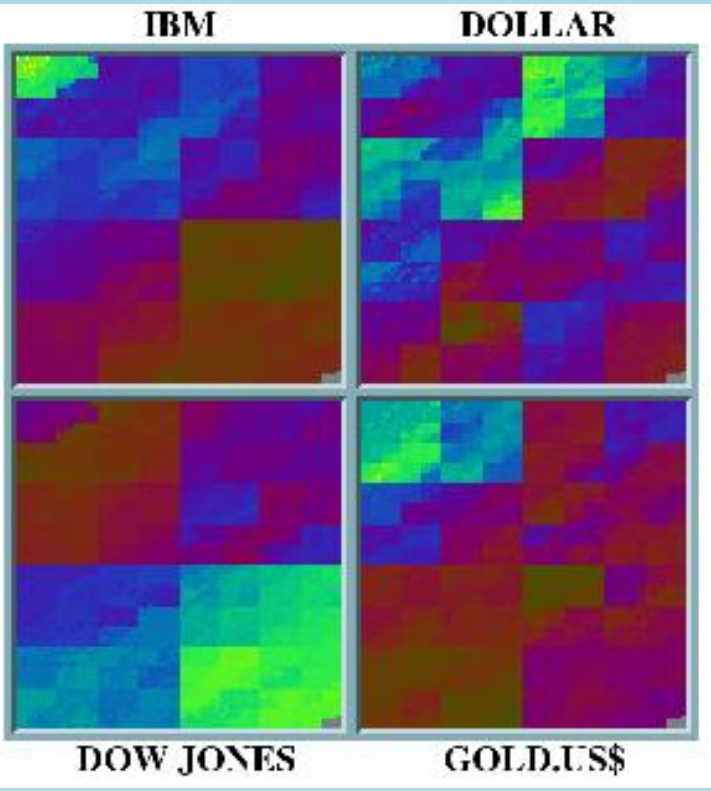
☞ Time series of stock data:



Line-by-line



Hilbert curve

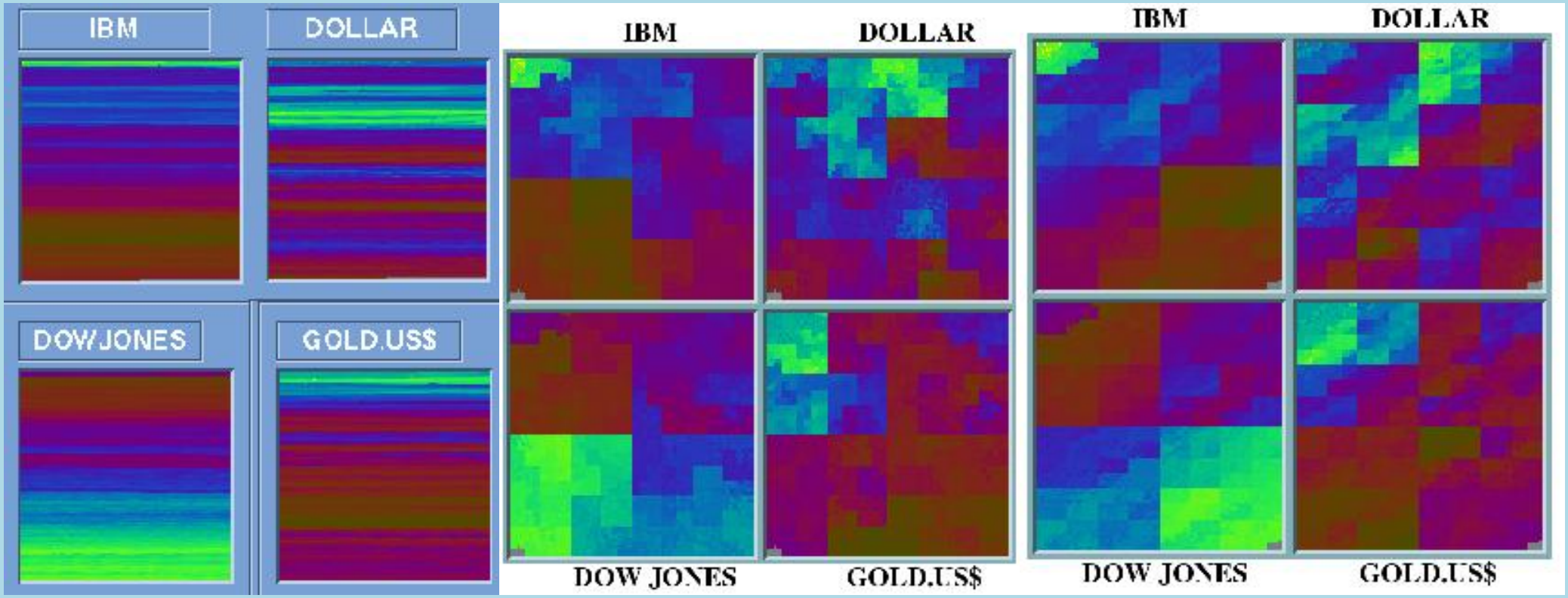


Z curve

Pictures are taken from “D. A. Keim, Databases and Visualization , Tutorial, Proc. ACM SIGMOD Int. Conf. on Management of Data, Montreal, Canada, 1996.”

Dense Pixel Displays (Example I)

☞ Time series of stock data:



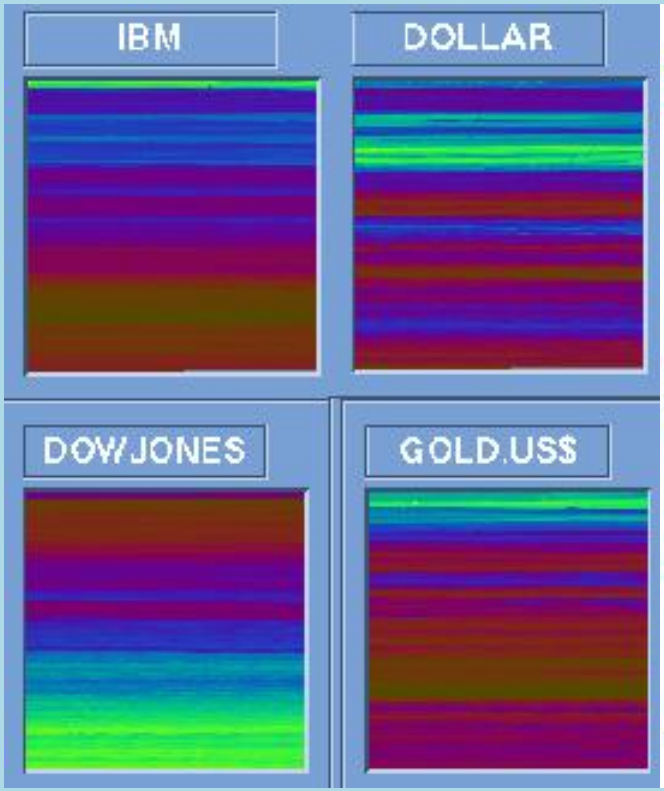
Line-by-line Hilbert curve Z curve

Pictures are taken from “D. A. Keim, Databases and Visualization , Tutorial, Proc. ACM SIGMOD Int. Conf. on Management of Data, Montreal, Canada, 1996.”

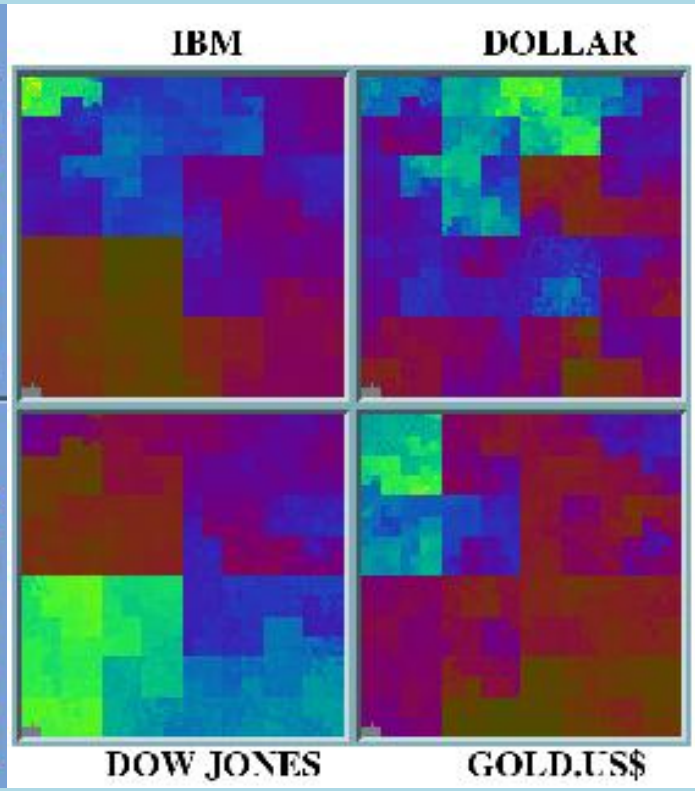
☞ Line-by-line order is good for time data

Dense Pixel Displays (Example I)

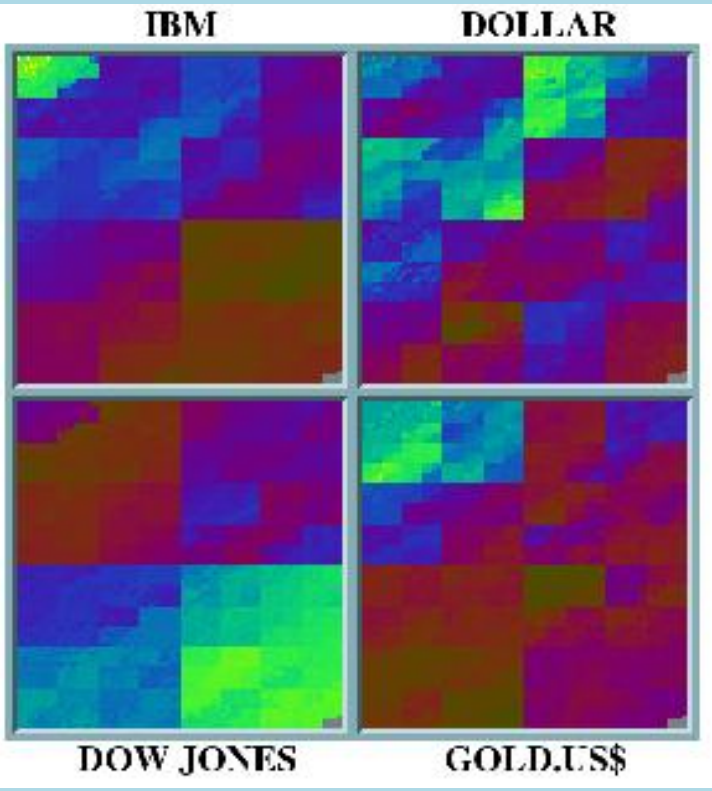
- Time series of stock data:



Line-by-line



Hilbert curve



Z curve

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- Line-by-line order is good for time data
- Space filling curve order is good for spatial data

Histograms (the Idea, Examples)

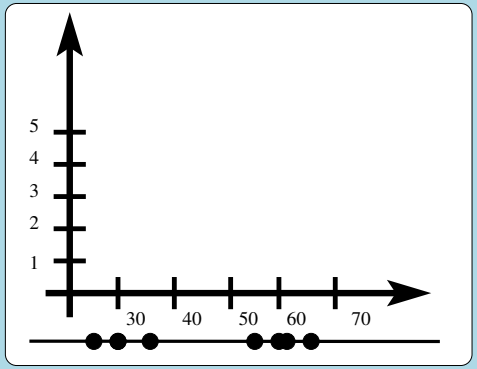
- ☞ There are two types of histograms:
  - Equi-height histograms
  - Equi-width histograms (quantiles)

Age	Salary	Height
35	1’500	1.58
65	2’300	1.88
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Histograms (the Idea, Examples)

- There are two types of histograms:
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  - Equi-width histograms (quantiles)
- The idea of equi-width histogram (1D case):
  1. Divide the domain into consequent buckets of size w.
  2. Count number of data points per each bucket
  3. The count is the height of the bucket

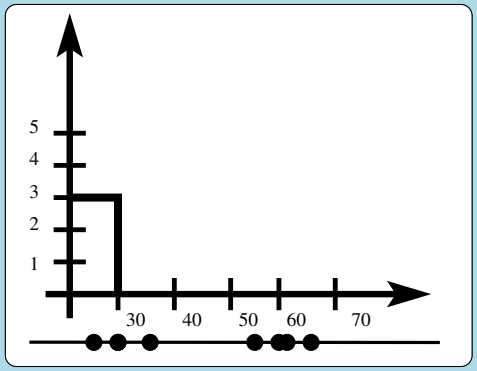
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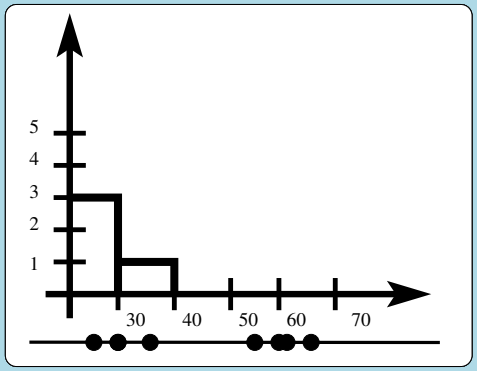
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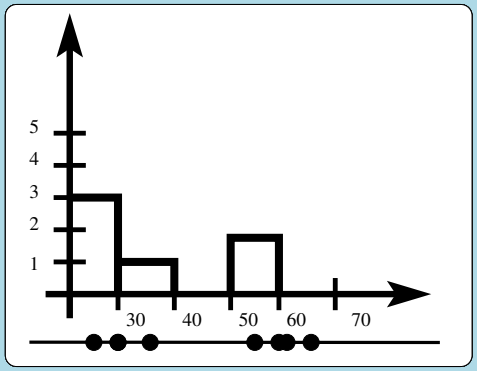
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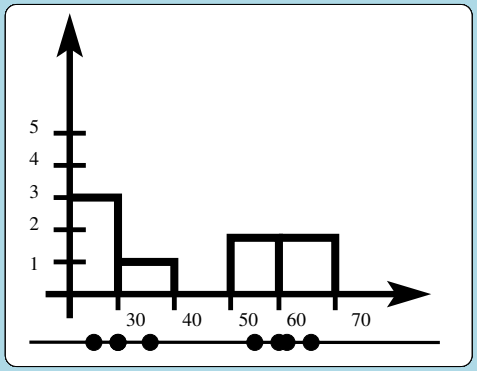
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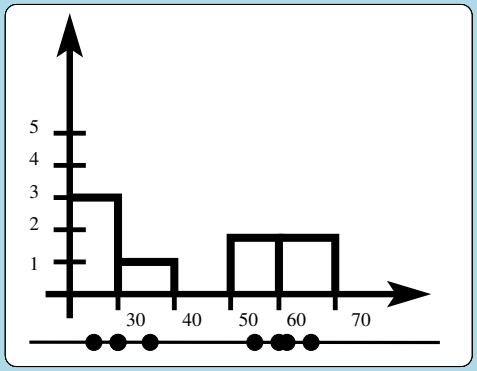
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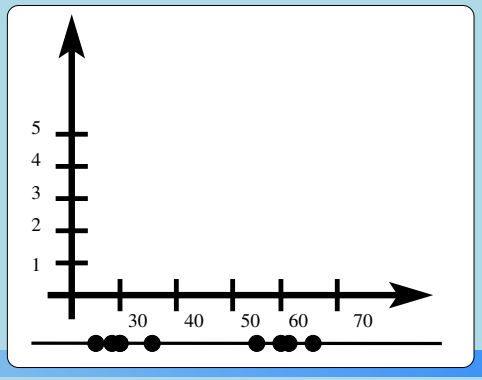


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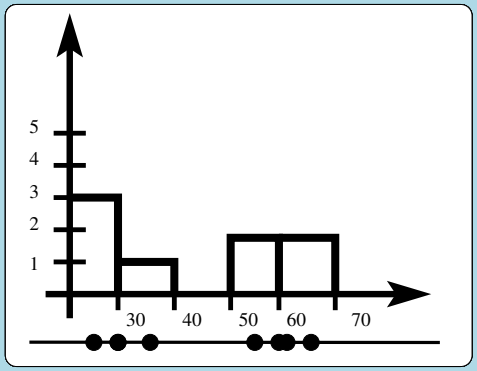


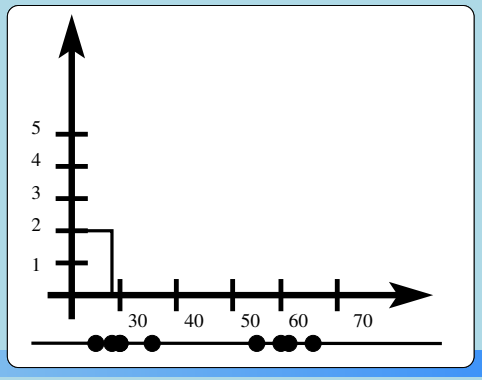


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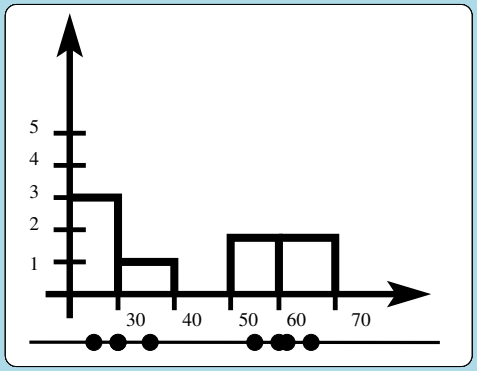


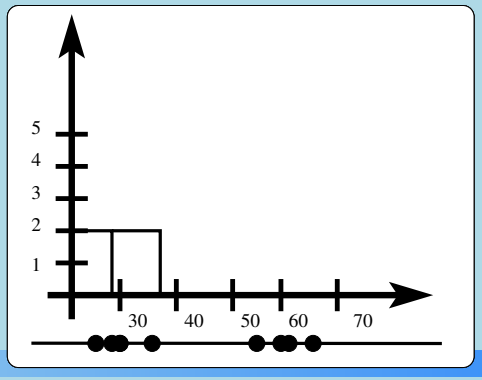


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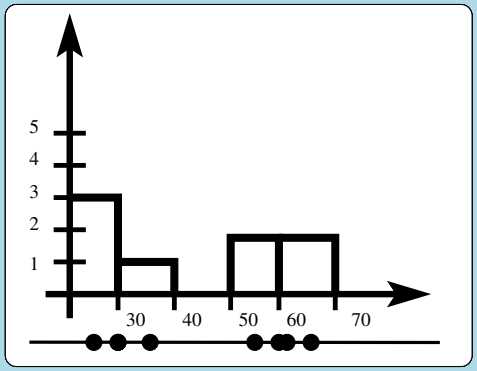


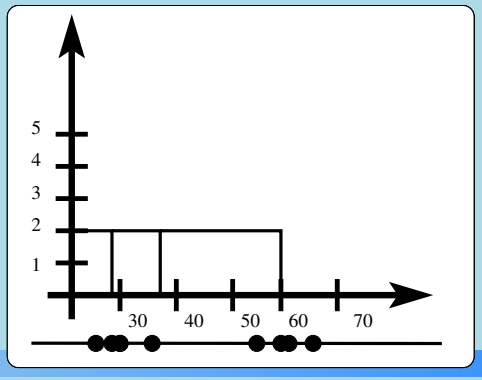


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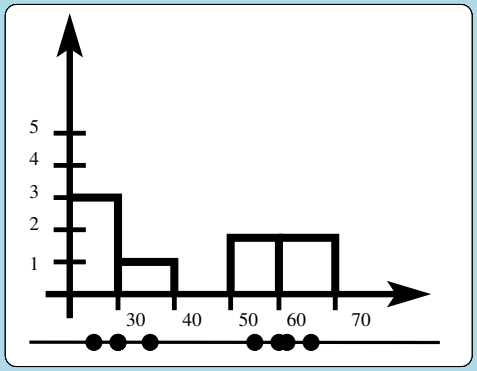


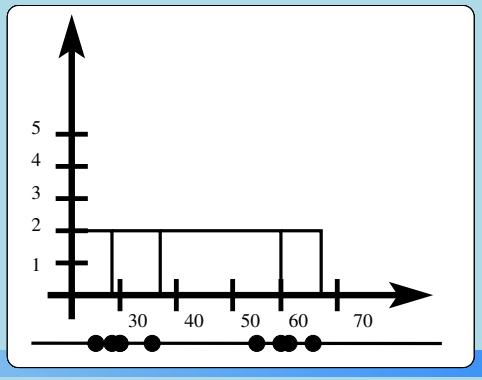


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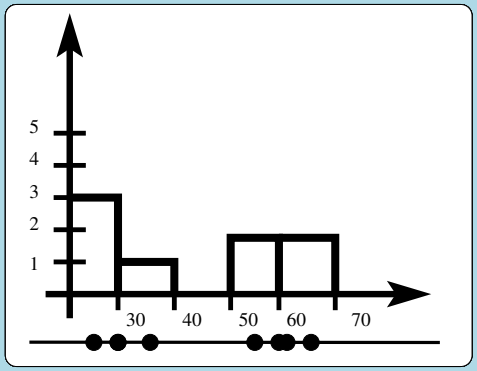


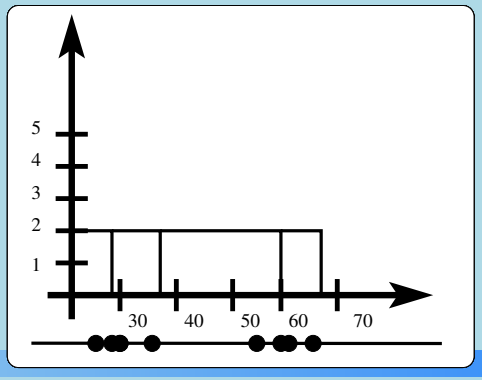


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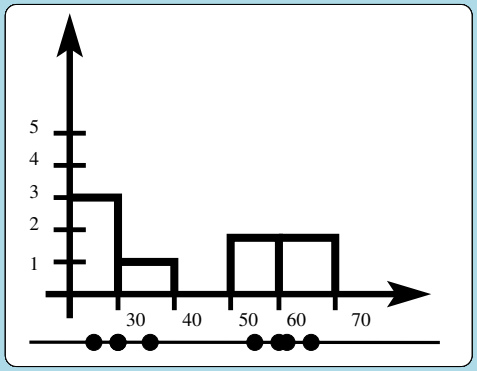


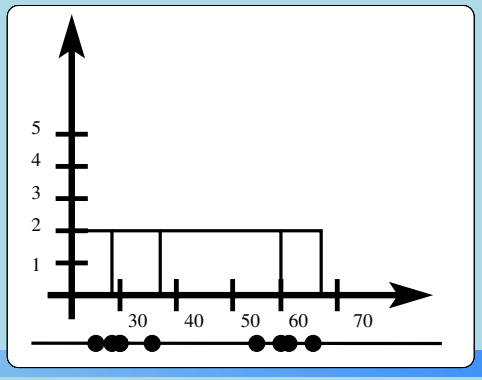


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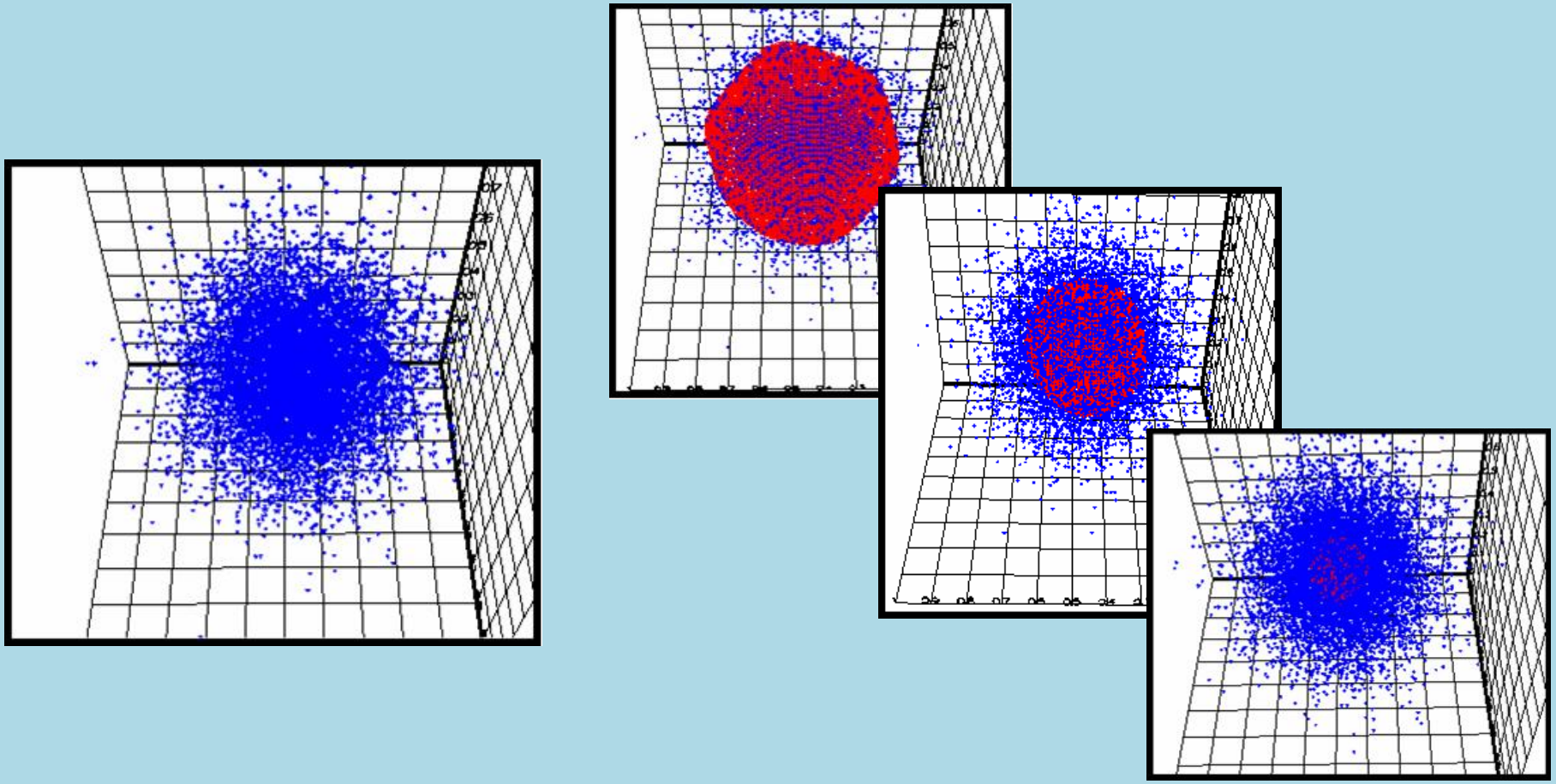
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- Exercise: equi-width histogram $w = 5$, $w = 30$

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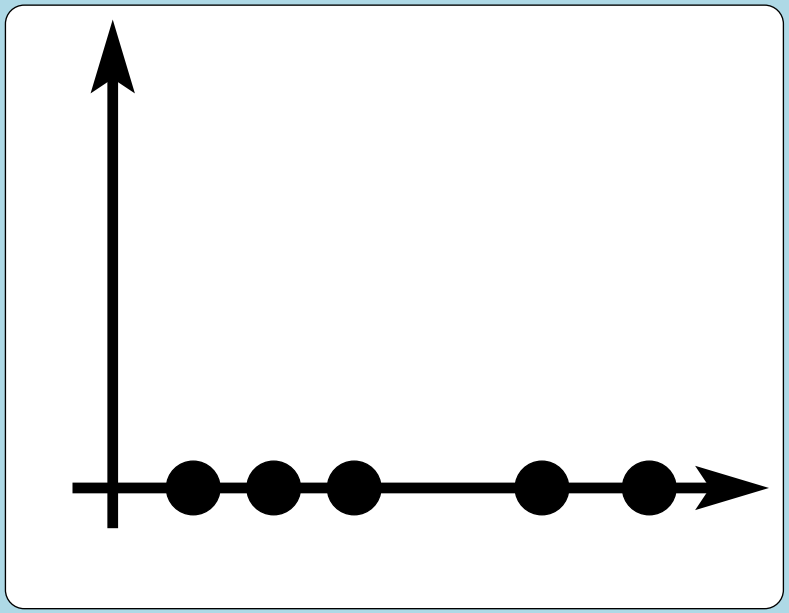
APDF, Density Surface Method (the Idea)



- We draw a surface, which encloses data points of α and higher density
  - First, we compute estimated probability density function (PDF) of the data
  - Second, we compute a density surface, based on the PDF

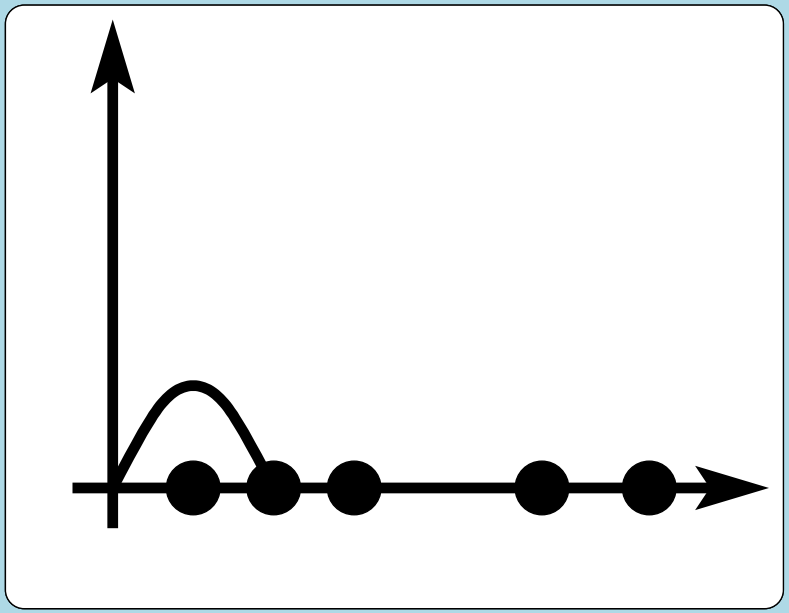
APDF, Density Surface Method (Estimation of PDF, Kernel Method, 1/3)

- PDF is an abstract mathematical model of the data. We use Kernel Method to estimate the mathematical abstraction
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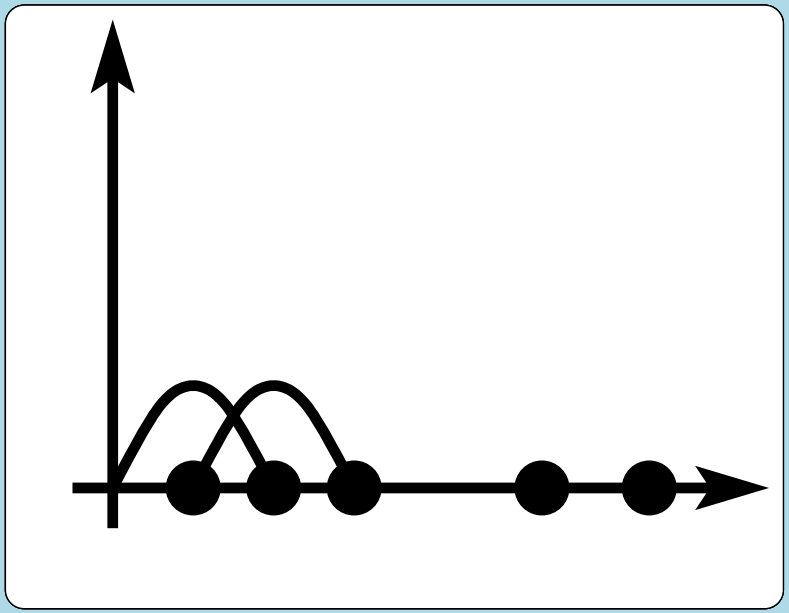
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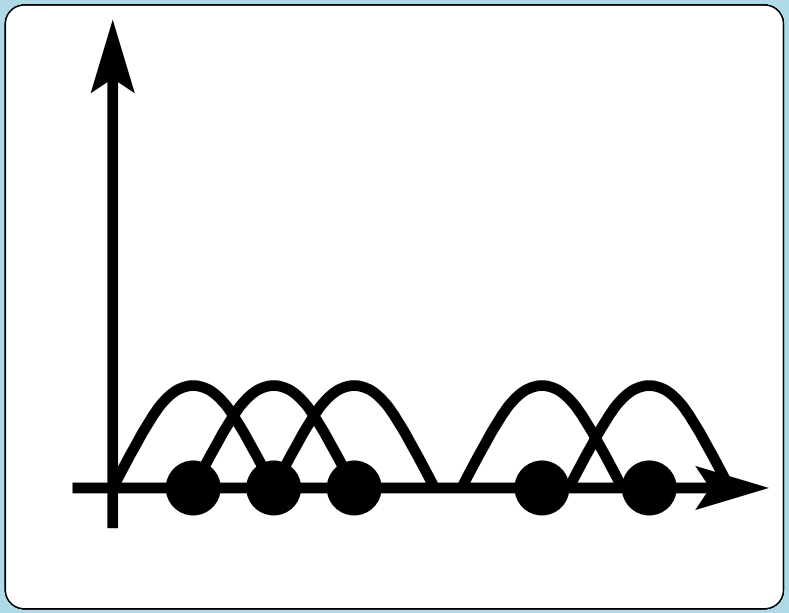
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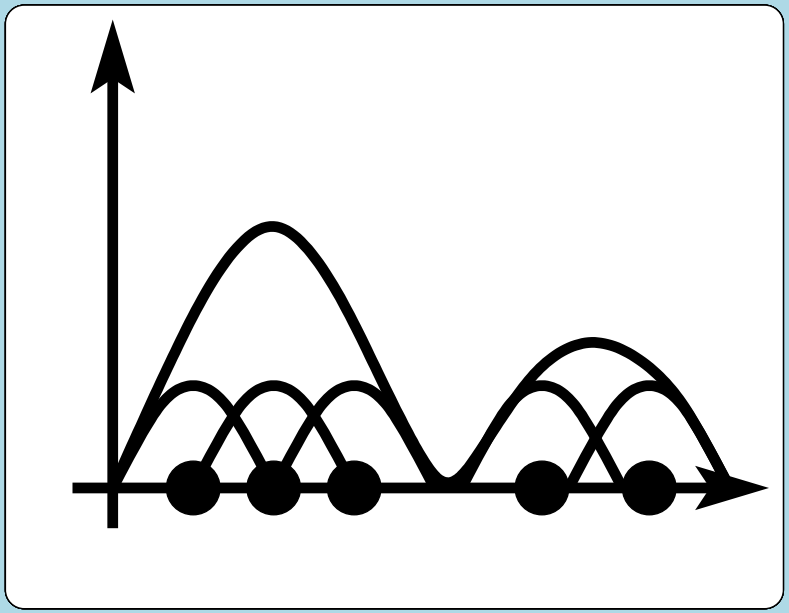
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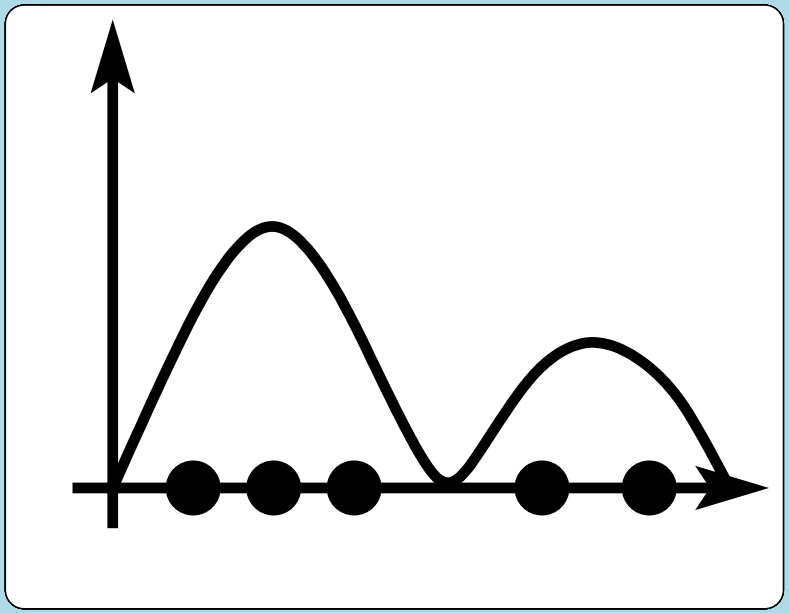
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- PDF reflects the density of the dataset: the value of the function is larger in denser regions of the dataset

APDF, Density Surface Method (Estimation of PDF, Kernel Method, 2/3)

- There are two parameters of the Kernel estimation:

APDF, Density Surface Method (Estimation of PDF, Kernel Method, 2/3)

- There are two parameters of the Kernel estimation:
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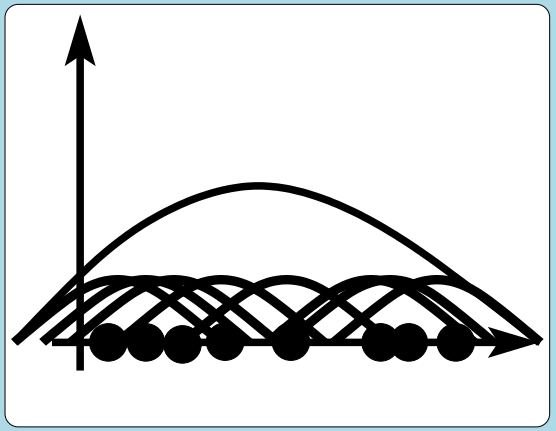
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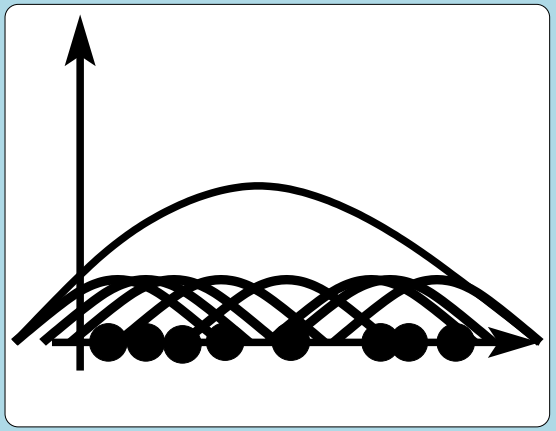
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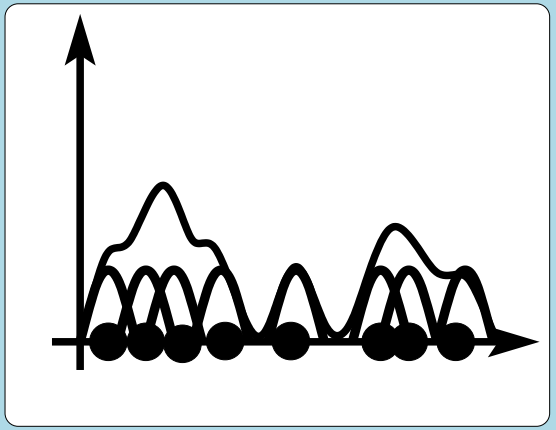


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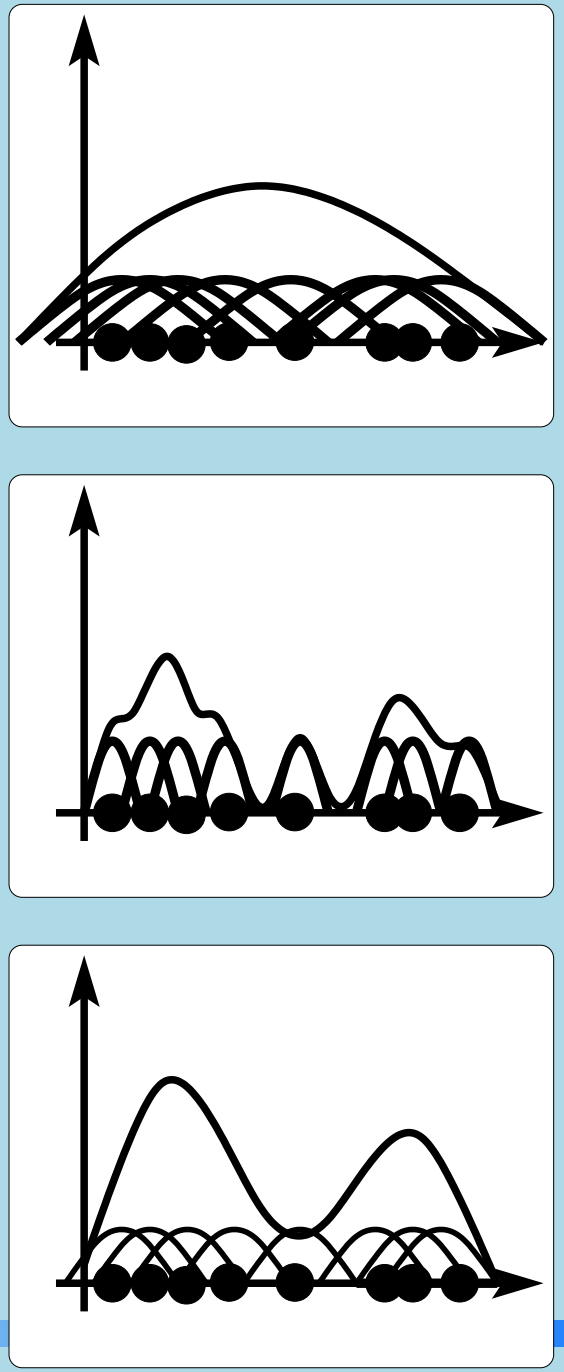
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- The smoothing parameter has a huge impact on the quality of the estimation:
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  - If the parameter is too small the PDF is undersmoothed
  - The right smoothing parameter depends on the size of database and (local) variance of the database.

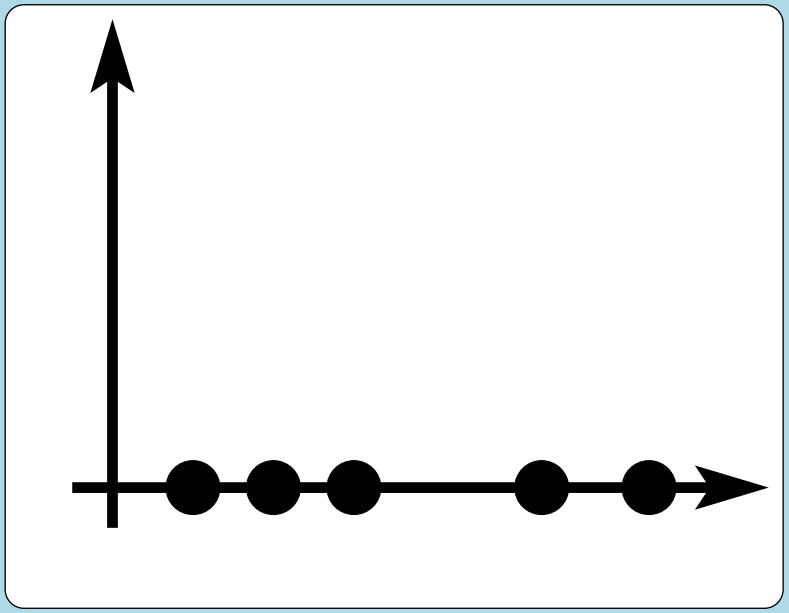


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☞ Mathematical background for 1D case:

- Say $D = \{d_1, d_2, \ldots, d_n\}$ is a given dataset. Kernel estimation of PDF for dataset D at point x is

(1)



☞ The smoothing parameter h is calculated based on the following equation:

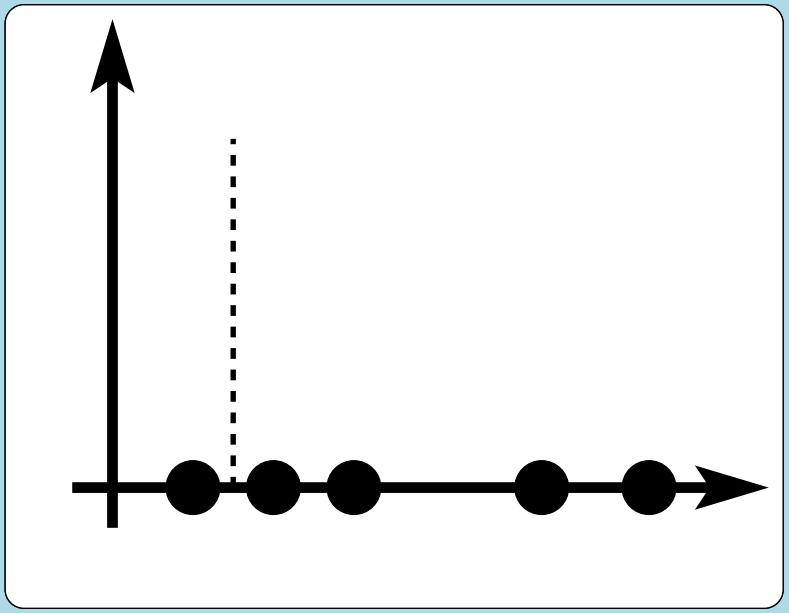
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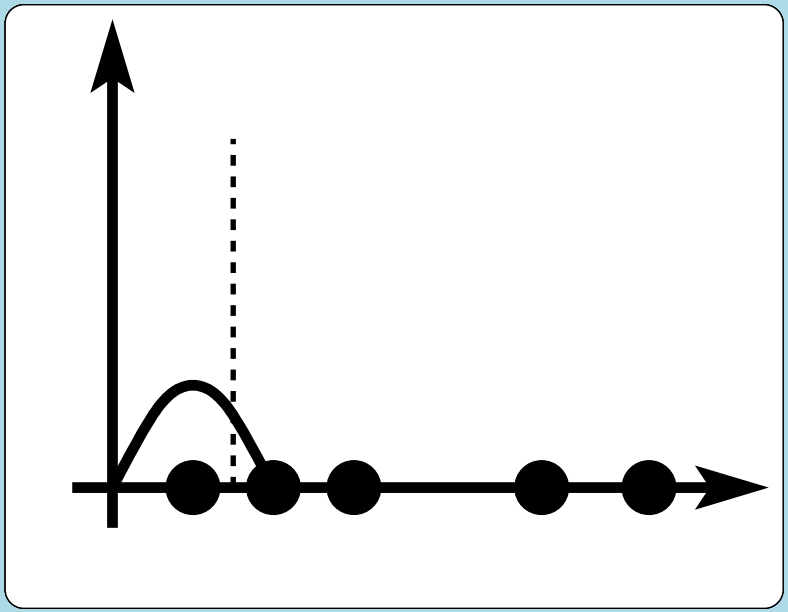
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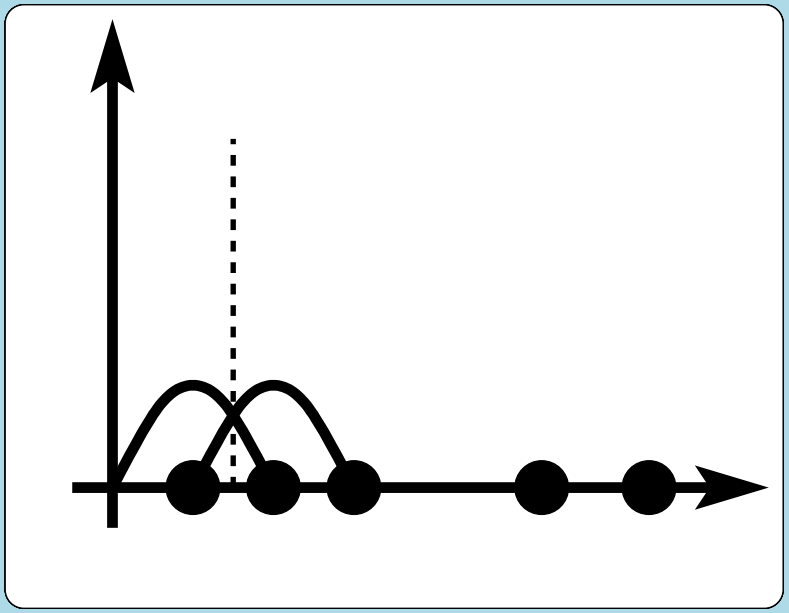
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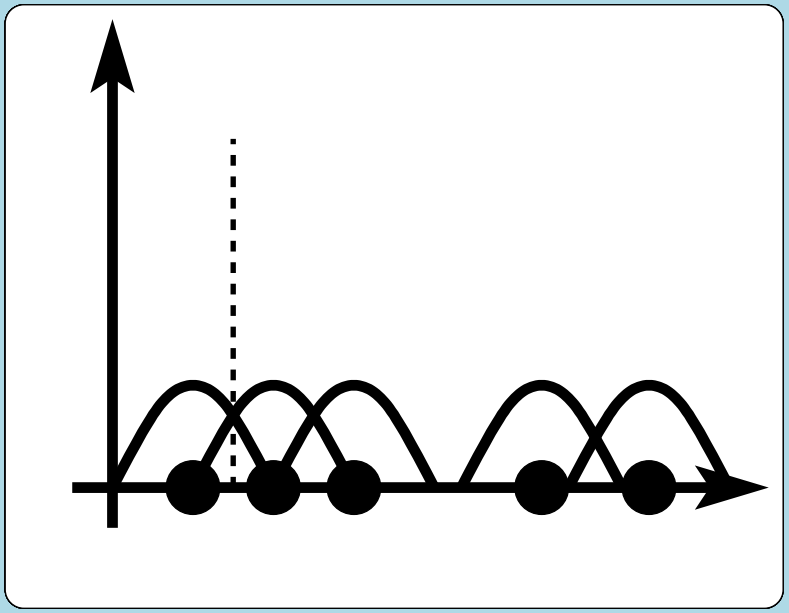
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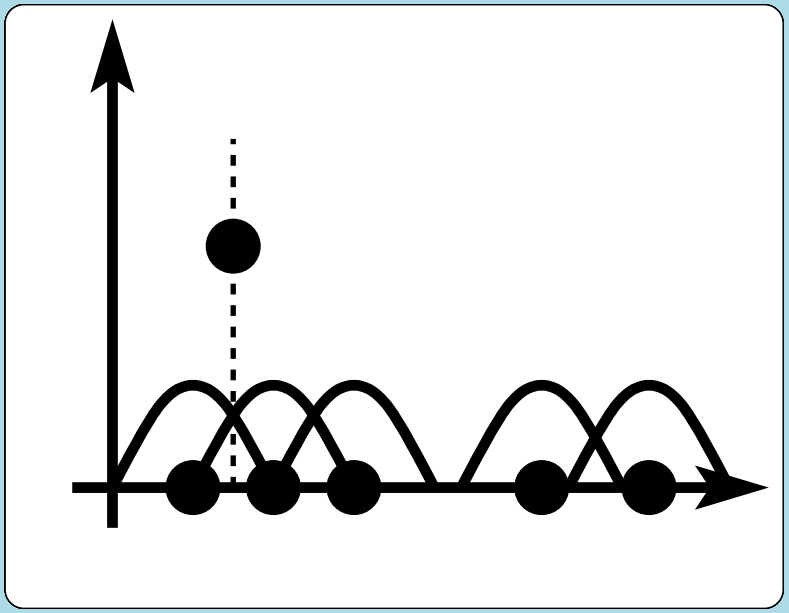
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$$\hat{f}_{K,h}(x) = \frac{1}{nh} K\left(\frac{|x - d_1|}{h}\right) + \frac{1}{nh} K\left(\frac{|x - d_2|}{h}\right) + \cdots + \frac{1}{nh} K\left(\frac{|x - d_n|}{h}\right) \quad (1)$$



☞ The smoothing parameter h is calculated based on the following equation:

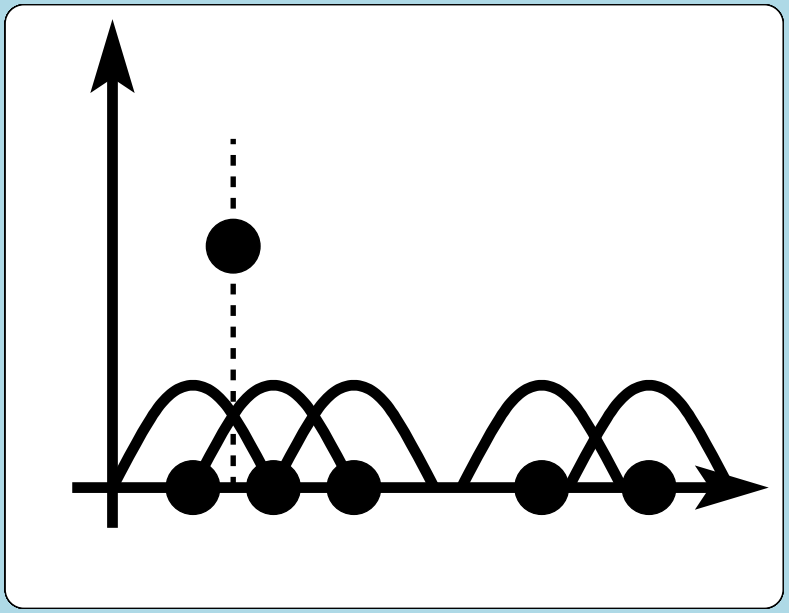
☞

APDF, Density Surface Method (Estimation of PDF, Kernel Method, 3/3)

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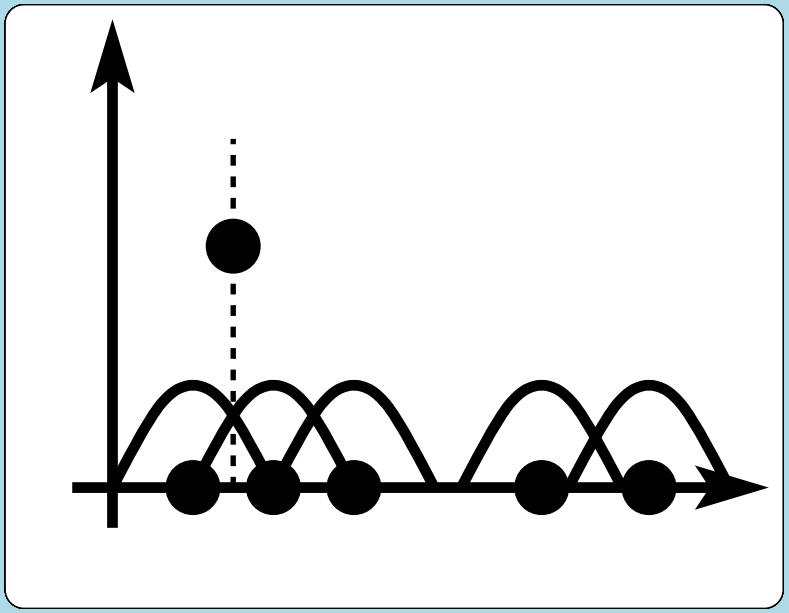
$$h = C(K)\sigma n^{-1/5} =$$

APDF, Density Surface Method (Estimation of PDF, Kernel Method, 3/3)

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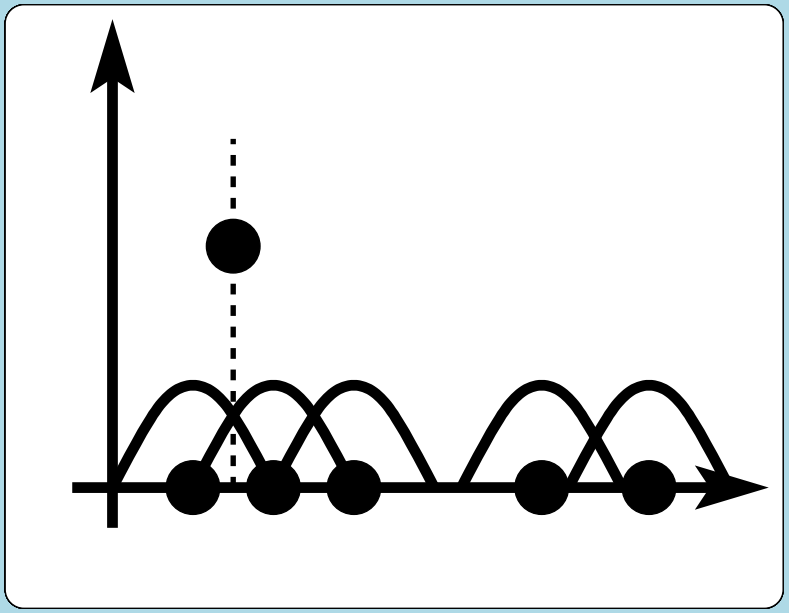
$$h = C(K)\sigma n^{-1/5} = C(K)\sqrt{\frac{1}{n}\sum_{i=1}^{n} d_i^2 - \Big(\frac{1}{n}\sum_{i=1}^{n} d_i^2\Big)^2}\, n^{-1/5}$$

APDF, Density Surface Method (Estimation of PDF, Kernel Method, 3/3)

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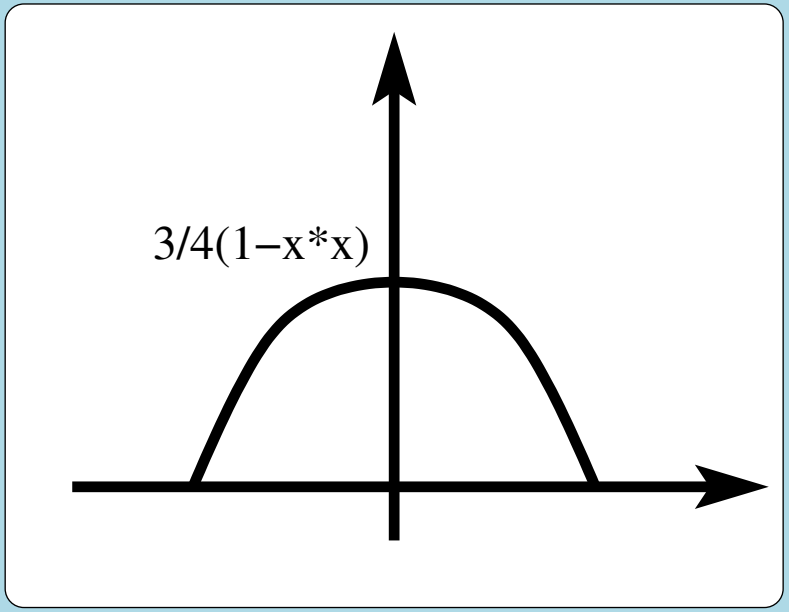
$$\hat{f}_{K,h}(x) = \frac{1}{nh} K\left(\frac{|x - d_1|}{h}\right) + \frac{1}{nh} K\left(\frac{|x - d_2|}{h}\right) + \cdots + \frac{1}{nh} K\left(\frac{|x - d_n|}{h}\right) \quad (1)$$



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☞ For Epanechnikov kernel:

$$h = 2.34\sigma n^{-1/5}$$

APDF, Density Surface Method (Estimation of PDF, Kernel Method, Example)

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- For Epanechnikov kernel:

$$h = 2.34\sigma n^{-1/5}$$

Age	Salary	Height
35	1'500	1.58
65	2'300	1.88
25	1'000	1.70
60	2'000	1.80
29	1'200	1.50
55	1'800	1.80
30	1'200	1.50
63	2'100	1.80

APDF, Density Surface Method (Estimation of PDF, Kernel Method, kD Case)

☞ Mathematical background for multivariate case:

- Say is a given dataset. Kernel estimation of PDF for dataset D at point is

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APDF, Density Surface Method (Estimation of PDF, Kernel Method, kD Case)

☞ Mathematical background for multivariate case:

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- $$h = C(K)\frac{1}{k}\sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_k^2}\, n^{-1/(d+4)}$$

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- For Epanechnikov kernel check B.W.Silverman “Density Estimation for Statistics and Data Analysis”, Chapman & Hall, 1985, p.p. 87.

APDF, Density Surface Method (Estimation of PDF, Kernel Method, kD Case)

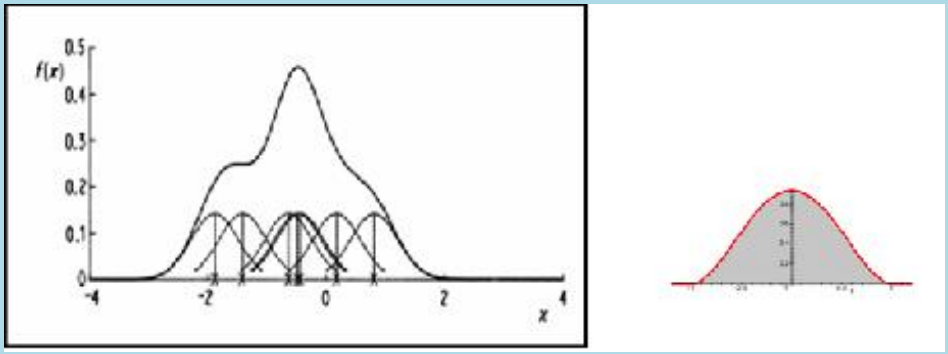
1D

2D

3D

APDF, Density Surface Method (Estimation of PDF, Kernel Method, kD Case)

1D

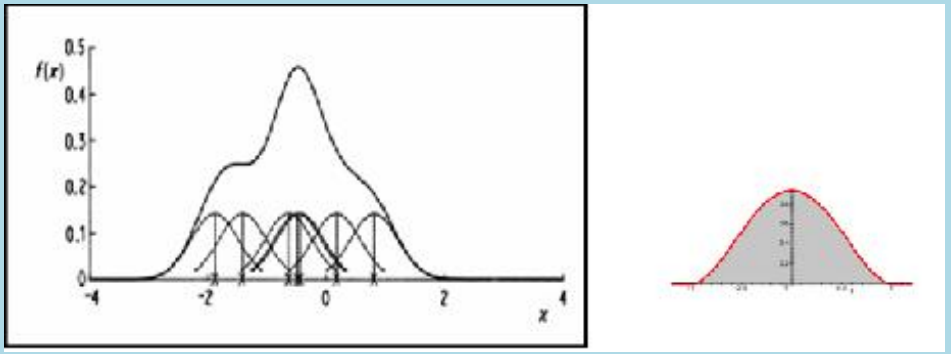


2D

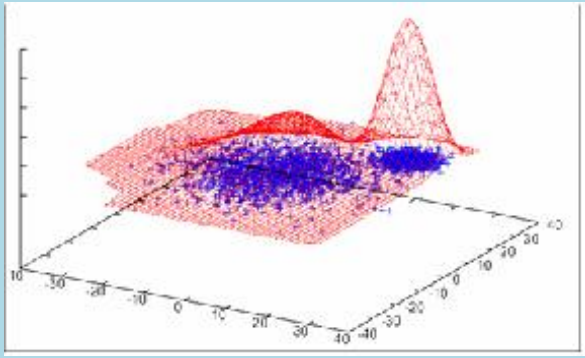
3D

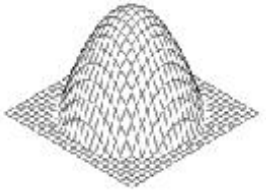
APDF, Density Surface Method (Estimation of PDF, Kernel Method, kD Case)

1D



2D

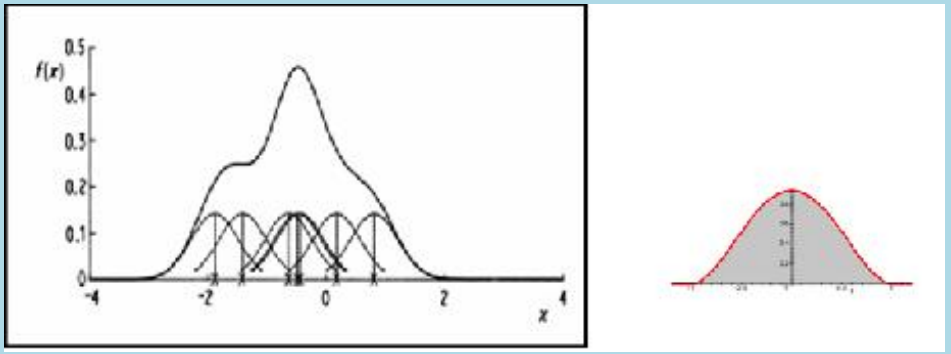




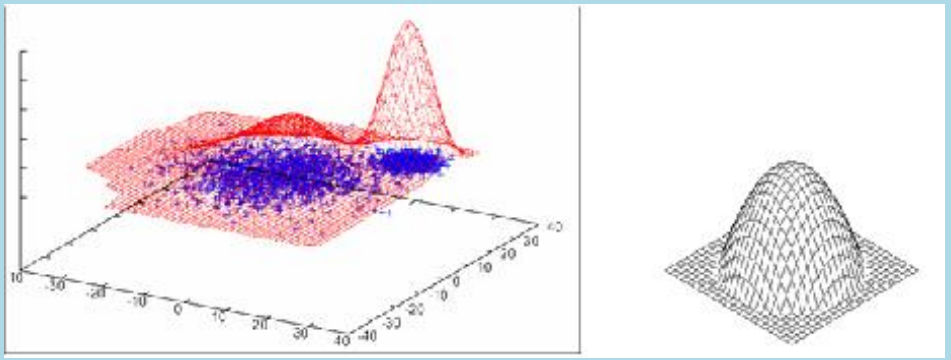
3D

APDF, Density Surface Method (Estimation of PDF, Kernel Method, kD Case)

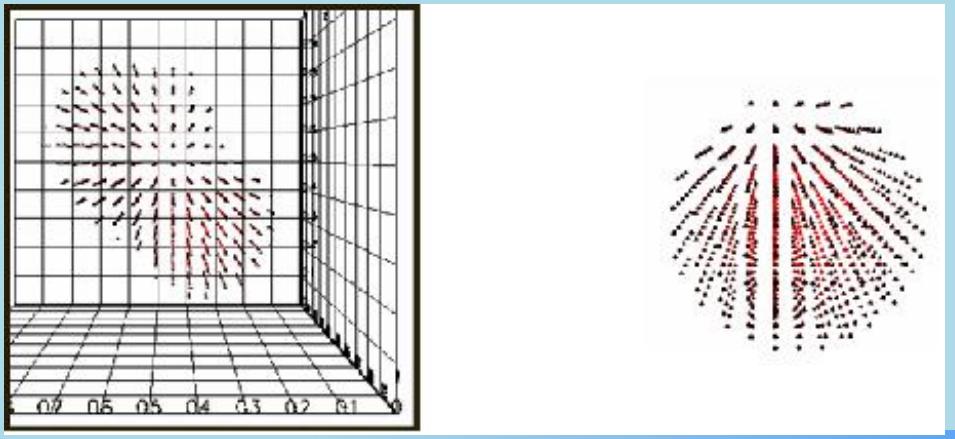
1D



2D



3D

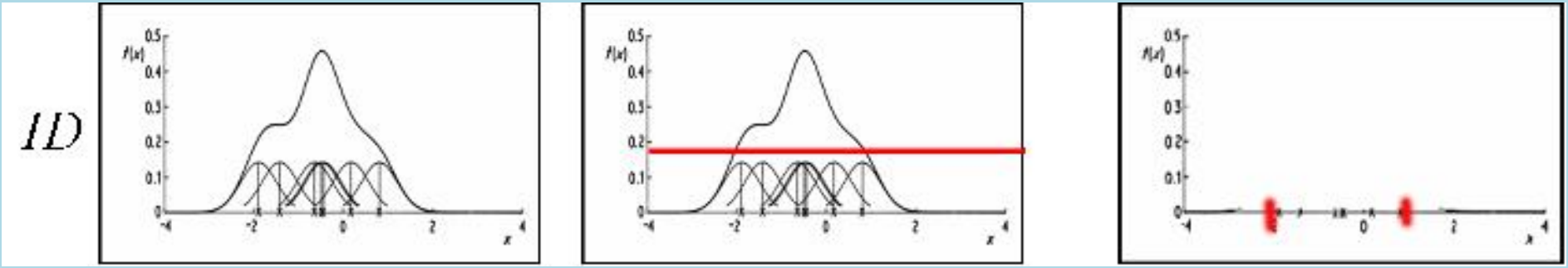


APDF, Density Surface Method (Calculation of DS, 1–2D Case)

- Density surface is a surface, which encloses data points of α and higher density
- A Density Surface is calculated by intersecting the PDF with a level hyper-plane

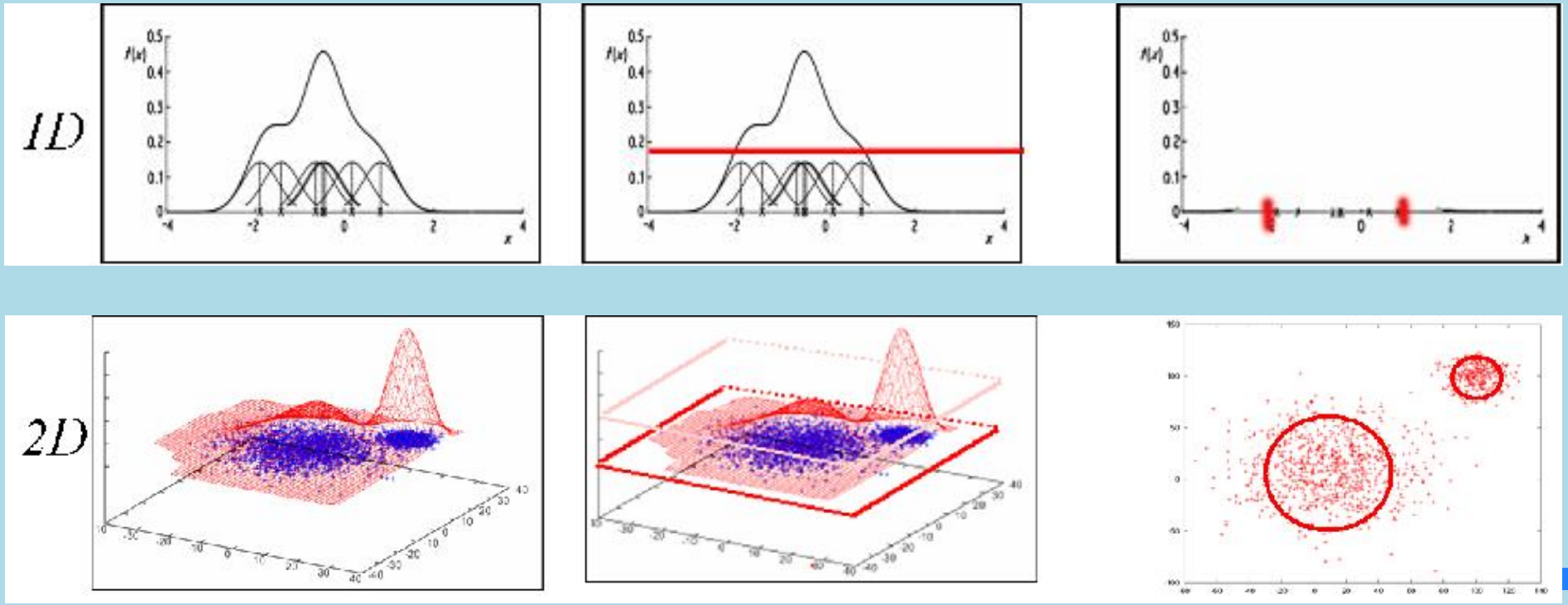
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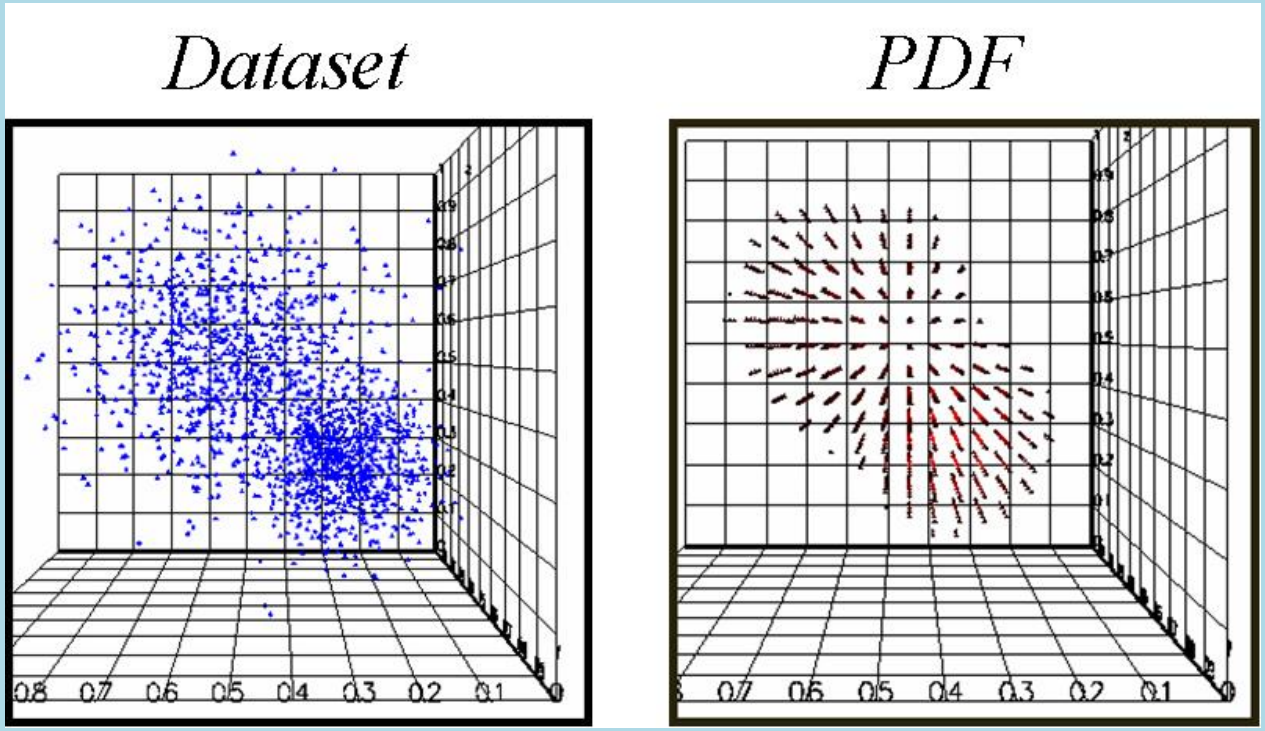


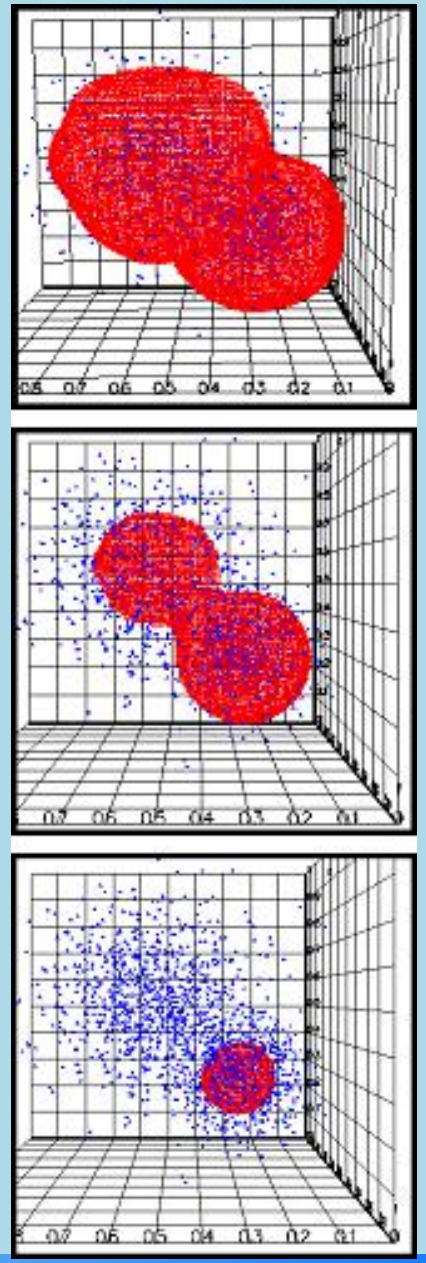
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APDF, Density Surface Method (Calculation of DS, 3D Case)

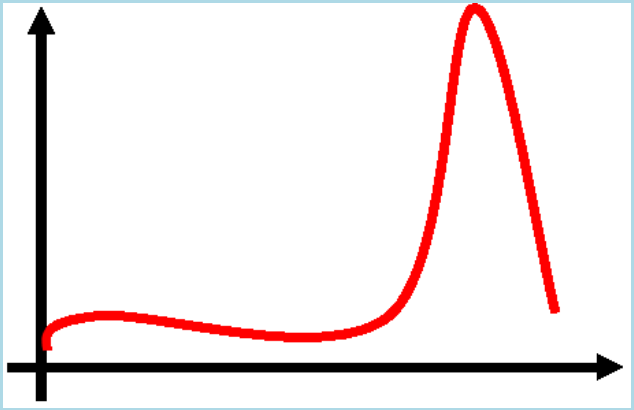




- Selecting a brightness level of the PDF yields a density surface
- Different density (pronounceness) complicates analysis of the data

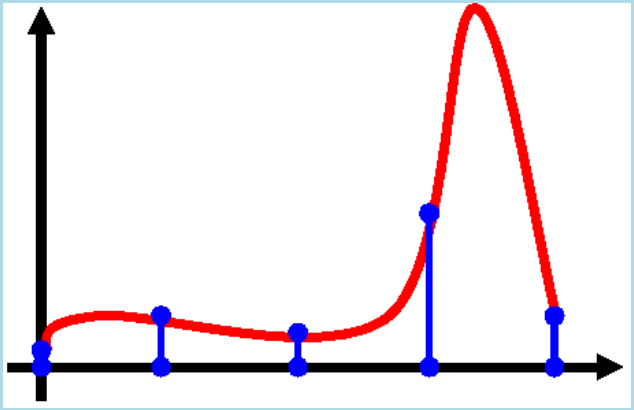
APDF, Density Surface Method (Implementation Aspects)

- PDF is implemented as a height unbalanced tree
  1. We start the estimation of the PDF at a uniform grid. The granularity of the cube is g
  2. We process a slice and calculate kernel additions at the grid points
  3. In the regions of non-linearity of the PDF we add more grids to describe the PDF
  4. Values of the PDF at new grid points we interpolate linearly
  5. We update the values with kernel additions of a next buffer
  6. We continue the steps 3–5 until the required precision is met



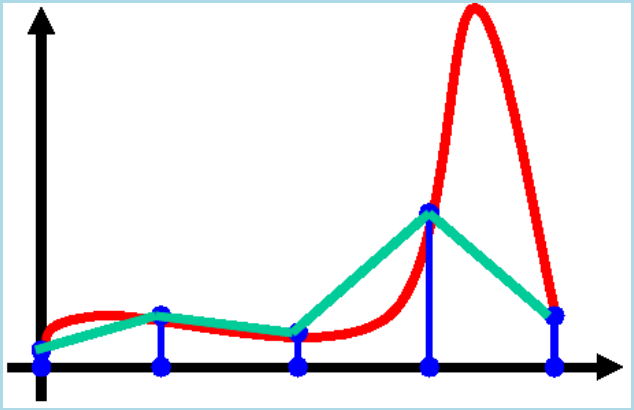
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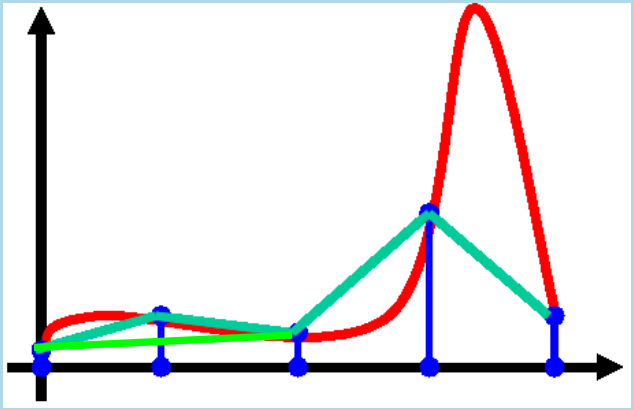
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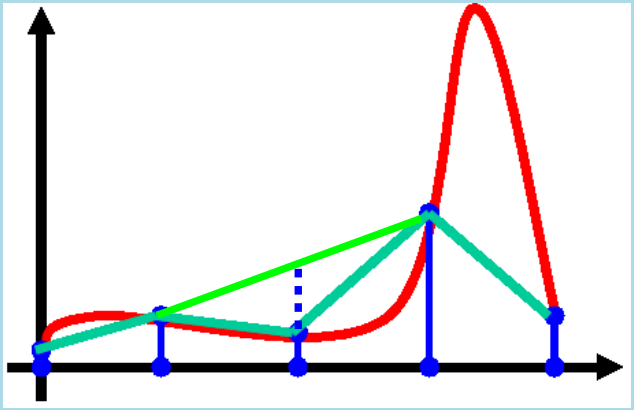
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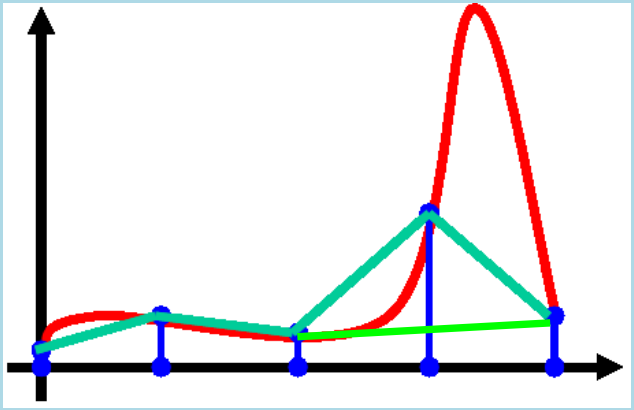
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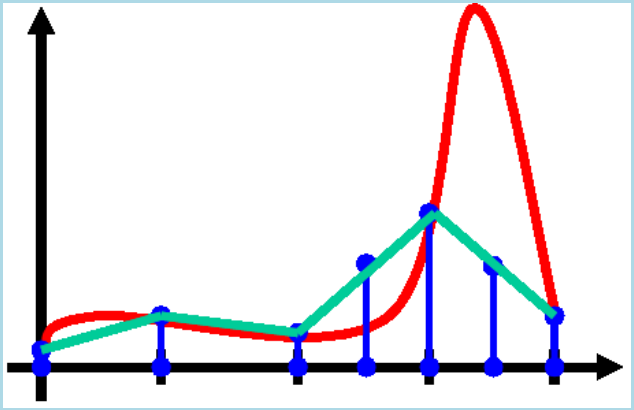
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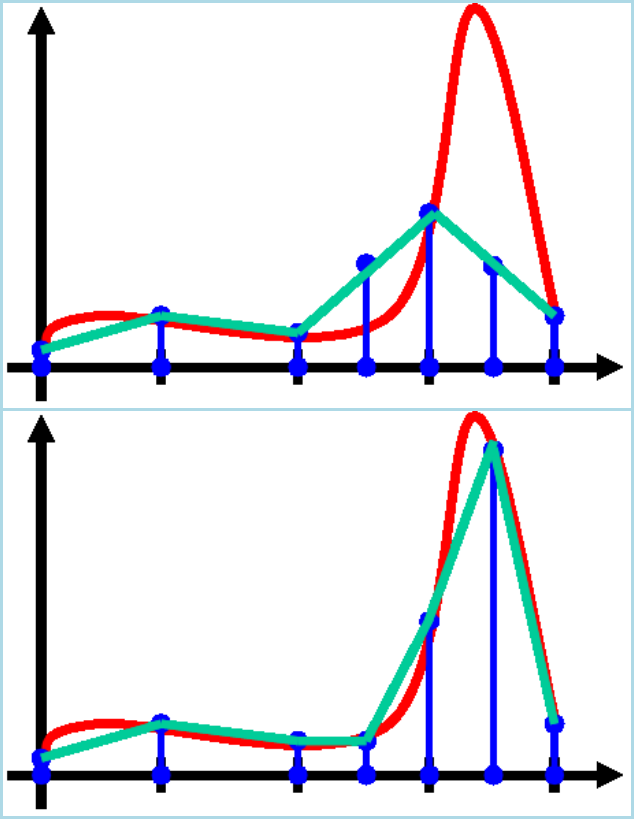
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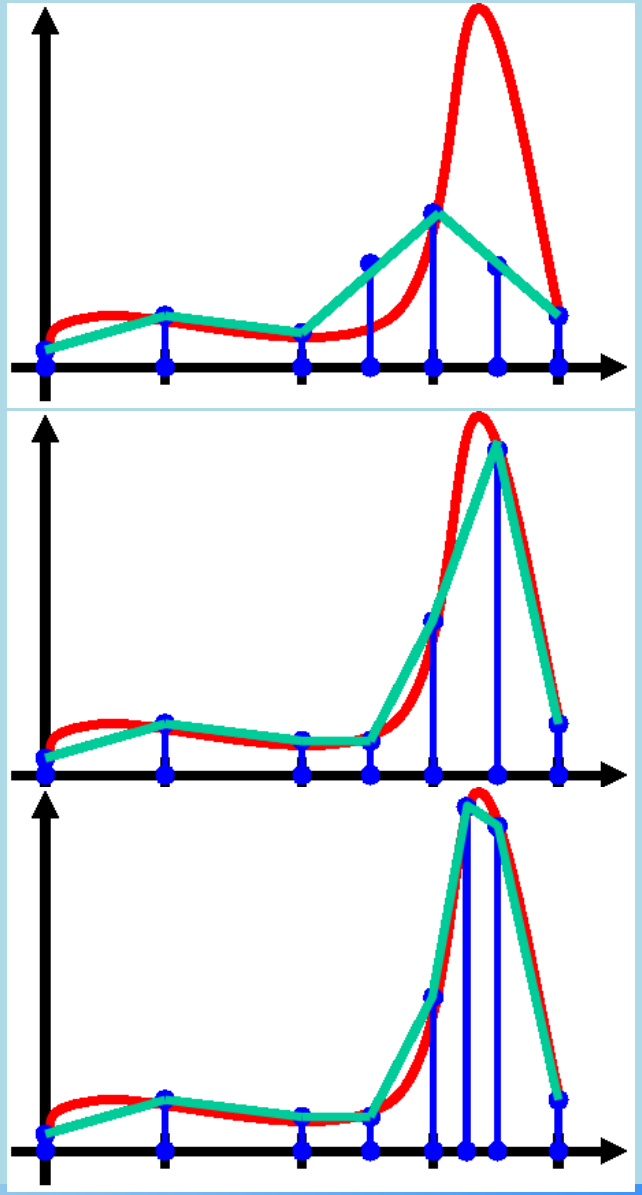
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  2. We process a slice and calculate kernel additions at the grid points
  3. In the regions of non-linearity of the PDF we add more grids to describe the PDF
  4. Values of the PDF at new grid points we interpolate linearly
  5. We update the values with kernel additions of a next buffer
  6. We continue the steps 3–5 until the required precision is met



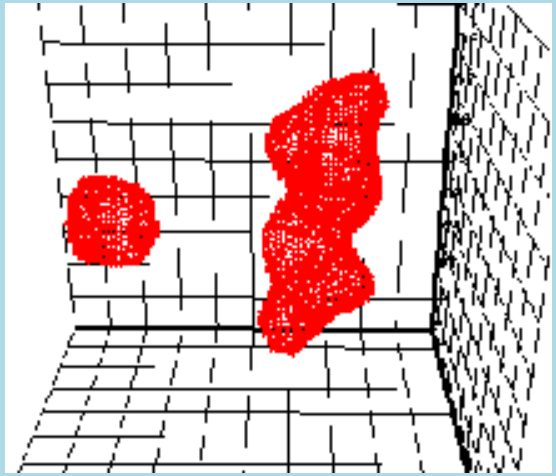
APDF, Density Surface Method (Implementation Aspects)

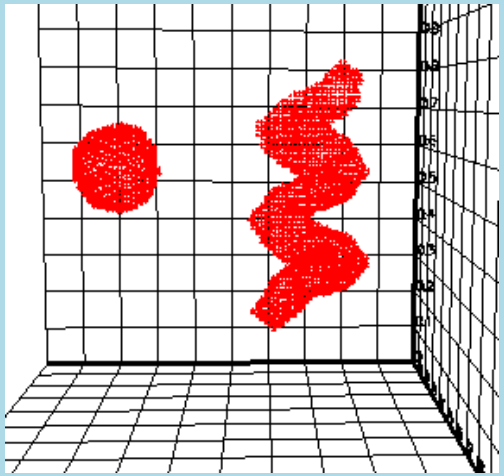
- PDF is implemented as a height unbalanced tree
  1. We start the estimation of the PDF at a uniform grid. The granularity of the cube is g
  2. We process a slice and calculate kernel additions at the grid points
  3. In the regions of non-linearity of the PDF we add more grids to describe the PDF
  4. Values of the PDF at new grid points we interpolate linearly
  5. We update the values with kernel additions of a next buffer
  6. We continue the steps 3–5 until the required precision is met

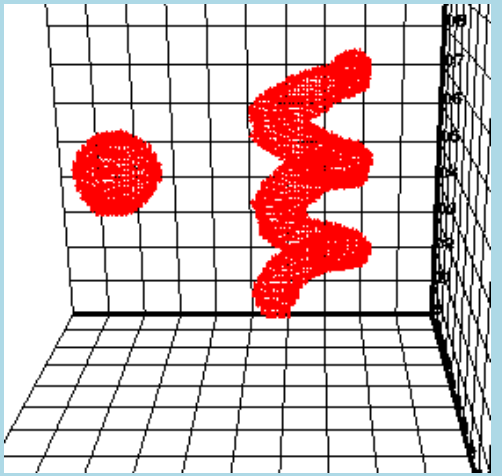


APDF, Density Surface Method (Empirical Evaluation)

- The experiments were run on PIII 1GZ, 0.5GB RAM, running Red-Hat Linux 7.2 OS
- It is feasible (but not easy!) to compute density surfaces interactively.
- A typical APDF tree is a few 100 KB.
- The typical computation time is a couple of seconds.

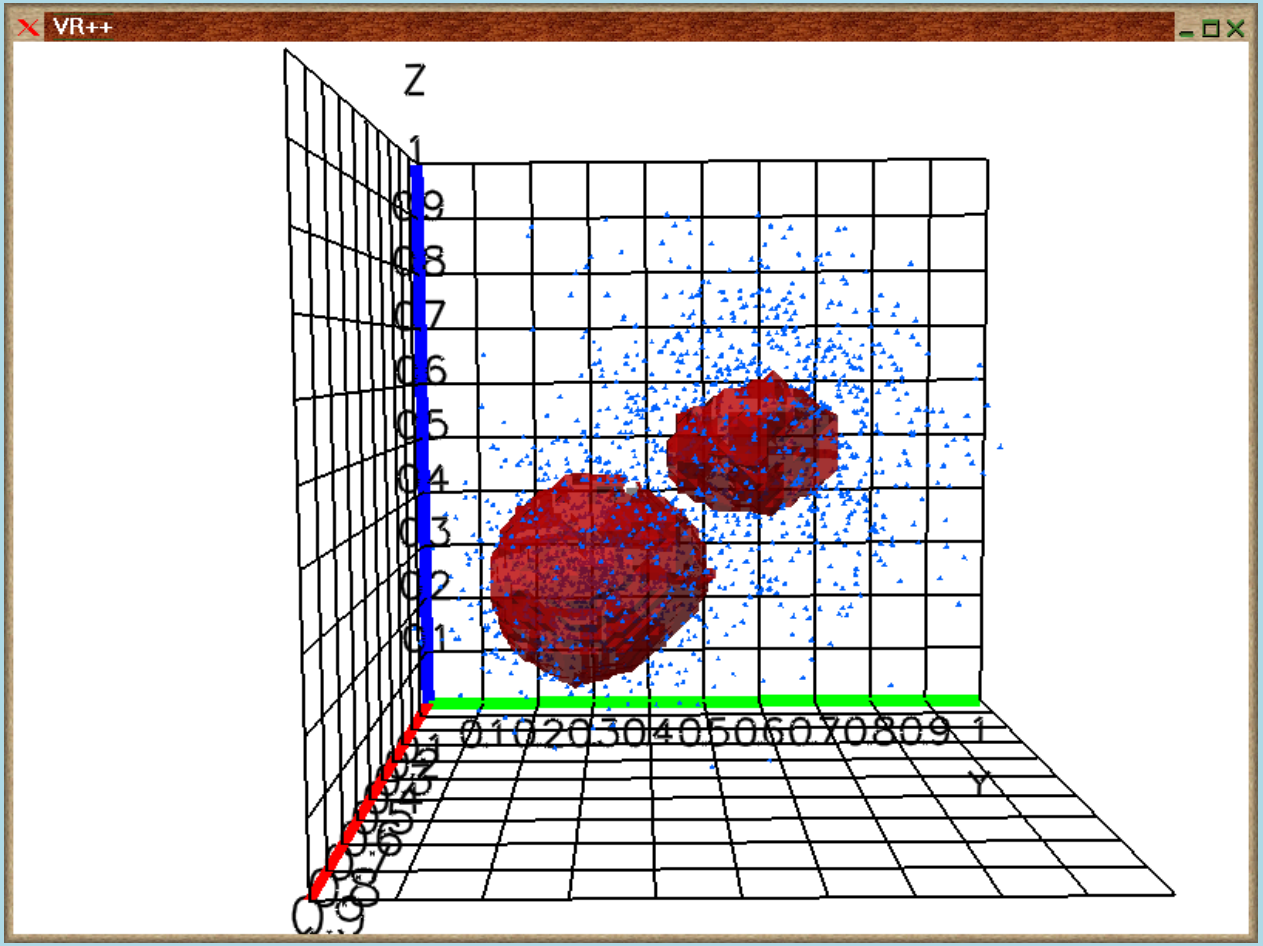






ε	0.9	0.5	0.1
Time	1.0 sec	2.6 sec	5.3 sec
Memory	173 KB	221 KB	257 KB

APDF, Density Surface Method (Demo of the 3DVDM System)



Demo of the VDM Systems (Required Packages)

- Make sure the following development packages are installed:
  - 3DVDM:
    - libtiff-devel
    - gtk-1.2-devel
    - lam-mpi
  - GGoby:
    - libxml2-devel

Demo of the VDM Systems (Downloading the Source)

- Downloading:
  - 3DVDM:
    - http://www.vrmedialab.dk/~vrpp/download/vr++-0.8.0.tar.gz
    - http://www.vrmedialab.dk/~vrpp/download/3dvdm-2.3.1.tar.gz
    - http://www.vrmedialab.dk/~vrpp/download/3dvdm_data.tar.gz
  - GGoby:
    - http://www.ggobi.org/Download/ggobi_1.0-0-beta.tar.gz

Demo of the VDM Systems (Untaring the source the Source)

- Untaring:
  - 3DVDM:
    - `$ tar xvfz vr++-0.8.0.tar.gz`
    - `$ tar xvfz 3dvdm-2.3.1.tar.gz`
    - `$ tar xvfz 3dvdm_data.tar.gz`
  - GGoby:
    - `$ tar xvfz ggobi_1.0-0-beta.tar.gz`

Demo of the VDM Systems (Compiling and Installing the Source)

- Compiling
  - 3DVDM:
    1. `$ cd vr++-0.8.0 && make && cd ..`
    2. `$ cd 3dvdm-2.3.1 && make && cd ..`
    3. If you get compilation errors, check `user.make` file and update include directories.
  - GGoby:
    - `$ cd ggobi && ./configure && make`
    - `$ make install` (under root)

Demo of the VDM Systems (Running the software)

- Running:
  - 3DVDM:
    2. `$ cd 3dvdm-2.3.1/bin`
    3. `$ ./ds` (or any other binary in the directory)
  - GGoby:
    - `$ /usr/local/bin/ggoby`